

SDK Reference Manual

CHCUSB DLL

(Windows XP / 7)

Function Specifications

Kodak Photo Printer 68XX

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Latest Edition:	April 1 st , 2025
Revision:	16.0

Revision History

Date	Revision	Description
November 17, 2011	10.0	Initial release.
August 17, 2015	12.0	Page 26, 27, 33 - Added "kPINFTAG_TABLEINFO" to chcusb_getPrinterInfo Page 31, 63, 67 - Revised Ribbon Codes of kPINFTAG_PAPER2 in chcusb_getPrinterInfo, and in chcusb_setPrinterToneCurve2 and chcusb_getPrinterToneCurve2. Page 29, 30, 40 <u>Added 6X6 size and 152x185mm size.</u> Page 28, 32 <u>Modified specification of below tag identifiers of chcusb_getPrinterInfo;</u> <u>kPINFTAG_PAPER</u> <u>kPINFTAG_PAPER2</u> Page 3, 7, 72 - Added chcusb_getColorSensorLevel.
April 3, 2019	13.0	Page 28-30, 32, 40 - Added print sizes as follows: 6x8 Combo2 (2x6-2up) 6x8 Combo1 (2x6-1up) 6x8 Combo2 (3x6-2up) 6x8 Combo1 (3x6-1up)
April 1 st , 2025	16.0	Updates based on YSD8-J0287E-16 Overall Document Corrections Page 1 - Added latest firmware versions for each printer type. Page 26, 28, 39 - Added the following tags to chcusb_getPrinterInfo: - kPINFTAG_ENERGY_SAVE_MODE_ENABLED - kPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME Page 40, 43, 44 - Added the following tags to chcusb_setPrinterInfo: - kPINFTAG_ENERGY_SAVE_MODE_ENABLED - kPINFTAG_ENERGY_SAVE_MODE_AWAKE - kPINFTAG_ENERGY_SAVE_MODE_SLEEP - kPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME Page 42 - chcusb_setPrinterInfo: - Revised the description of kPINFTAG_POLISH Page 68 / 70 - chcusb_getPrinterToneCurve / chcusb_getPrinterToneCurve2 - Removed kGetPrinterTCTbl from the description of type.

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1. SCOPE

This manual specifies a CHCUSB DLL of a built-in-application type printer driver for the KODAK 6800 and KODAK 6850 Photo Printer (hereafter referred to as the printer unless otherwise specified).

Model	Applicable Firmware Version
6800 Type 1	Version 261
6800 Type 2	Version 361
6800 Type 3	Version 461
6850	Version 679 or Version 879

The KODAK 6800 and KODAK 6850 Photo Printer software development kit, programmer's guide, and sample application executable and source code are intended as a development aid and example of usage for software engineers wishing to develop their own applications for the KODAK 6800 Photo Printer and/or the KODAK 6850 Photo Printer.

No warranty of any kind is expressed or implied, including, but not limited to, warranties of merchantability and fitness for a particular purpose. While the code may have undergone limited testing, Kodak Alaris does not warrant that the software or documentation will be error free, or that defects will be corrected. Use of the sample code, or implementation of any techniques described, is done so at the user's own risk and it is the sole responsibility of the user to ensure that the code performs as it should.

2. OUTLINE OF THE PRODUCT

This CHCUSB DLL is an accessory program that provides a function to transfer image data to an applicable printer from the application program. This product works as a DLL module that can be built in an application program. But the image transmit format should be BGR point sequential raw data of 8bit type.

3. SOFTWARE CONFIGURATION

3.1 Program Configuration

This CHCUSB DLL consists of the four files listed below. These files are used by arranging them in the same directory as the program executable file (.exe file) of an application.

- CHCUSB.dll : Main body of the CHCUSB DLL
- SDKColorMatch.dll : CHCUSB submodule for color matching
- SDKUsbCtrl.dll : USB access module

3.2 Development Kit Configuration

A development kit for the CHCUSB DLL contains the following components:

- DLL files (CHCUSB.dll, SDKColorMatch.dll, SDKUsbCtrl.dll)
- C header file (Chcusbif.h)
- CHCUSB.DLL import library file (CHCUSB.LIB)
- Sample Codes

4. SYSTEM REQUIREMENTS

4.1 Host Computer Requirements

4.1.1 CPU

The CHCUSB DLL module can run on an IBM-PC/AT compatible computer that runs any of the operating systems listed in 4.1.2.

4.1.2 Operating system (OS)

This CHCUSB DLL module is available for the following operating systems:

- Microsoft Windows 98 *
- Microsoft Windows Me *
- * Need to install the Microsoft USB print driver.
- Microsoft Windows XP Professional/HomeEdition.
- Microsoft Windows XP Professional x64 Edition (running as 32-bit under WOW64 emulation).

- Microsoft Windows Windows 2003 Server.
- Microsoft Windows Vista.
- Microsoft Windows Vista x64 (running as 32-bit under WOW64 emulation).
- Microsoft Windows 7 x86 (32-bit).
- Microsoft Windows 7 x64 (running as 32-bit under WOW64 emulation).

4.1.3 Interface

Complies with USB 2.0.

4.2 Application Requirements

The application program must generate point-sequential RGB bitmap image data. Text and vector files must be rasterized in advance into RGB bitmap image data.

5. DEVELOPMENT ENVIRONMENT

The CHCUSB DLL was developed in the following environment:

- Compiler : Microsoft Visual C++ v6.0
- Operating System : Operating Systems listed in 4.1.2.

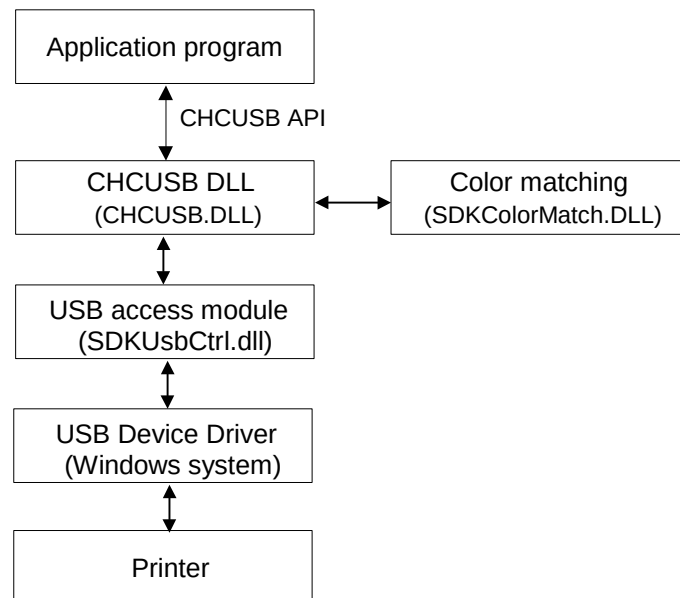
6. ASSEMBLING INTO APPLICATIONS

6.1 Module Configuration

The main body of the CHCUSB DLL (CHCUSB.dll) provides an application program interface (CHCUSB API) for controlling a printer by the application program.

The color matching module (SDKColorMatch.dll) is used with the CHCUSB DLL (CHCUSB.dll) to allow use of an ICC profile for color matching.

The CHCUSB DLL usage is shown below.



6.2 Functions Exported by DLL

The CHCUSB DLL exports a function in the following format:

- For C processing systems
function-type APIENTRY function-name ()
Example: BOOL APIENTRY chcusb_open (WORD*rResult);

The CHCUSB DLL exports the functions listed below.

- For C processing systems

Function	Export number (Sequence value)	Function
chcusb_makeThread	1	Defined maximum thread number of Multi-Thread.
chcusb_open	2	Initializes the CHCUSB DLL and generates a new thread.
chcusb_close	3	Closes the current thread, and closes the work memory and DLLs that the CHCUSB DLL uses.
chcusb_releaseThread	4	Closes all threads, and closes the work memory and DLLs that the CHCUSB DLL uses.
chcusb_listupPrinter	5	Lists the printer IDs of the printers that are connected to the computer.
chcusb_listupPrinterSN	6	Lists the serial numbers of the printers that are connected to the computer.
chcusb_selectPrinter	7	Selects a printer (using printer ID) that the computer uses.
chcusb_selectPrinterSN	8	Selects a printer (using printer serial number) that the computer uses.
chcusb_getPrinterInfo	9	Gets the firmware version of the selected printer, print count, and USB information.
chcusb_imageformat	10	Sets an image size and a transfer mode.
chcusb_setmtf	11	Sets parameters for profile emphasis and smoothing.
chcusb_makeGamma	12	Creates a tone curve table.
chcusb_setlcctable	13	Specifies a color matching table and an I/O tone curve.
chcusb_copies	14	Sets the number of copies to be printed.
chcusb_status	15	Checks the status of the currently-selected printer.
chcusb_statusAll	16	Checks the status of all printers connected to the machine.
chcusb_startpage	17	Starts page processing and sends an image header and print parameters.
chcusb_endpage	18	Ends page processing.
chcusb_write	19	Sends image data.
chcusb_setPrinterInfo	20	Sets parameters such as media types for the printer.
chcusb_getGamma	21	Reads the tone curve table from the file.
chcusb_getMtf	22	Reads the MTF file from the file.
chcusb_cancelCopies	23	Cancels the print count setting for the image which is being printed or is to be printed next and makes the printer ready when the printout of the current page ends.
chcusb_setPrinterToneCurve	24	Changes tone curve inside printer.
chcusb_getPrinterToneCurve	25	Acquires tone curve inside printer.
chcusb_blinkLED	26	Flashes LED of selected printer.
chcusb_resetPrinter	27	Resets selected printer.
chcusb_attachThreadCount	28	Checks the thread count of the used and maximum set.
chcusb_getColorSensorLevel	29	Acquires color sensor Level of the ribbon set to the printer.

6.3 Assembling the DLL

6.3.1 Linking the IMPORT library of the DLL

The functions listed in 6.2 are available when the IMPORT library (for Visual C++) of the CHCUSB DLL (CHCUSB.dll) is added to a Visual C++ project.

The CHCUSB.dll is loaded when an application program is executed and discarded when the application program ends.

The DLL file to be loaded is searched in the following order of priority:

- (1) Directory loading the application program
- (2) Current directory
- (3) System directory of the 32-bit Windows OS
The path of this directory can be obtained by the "GetSystemDirectory" function. The directory name is SYSTEM32.
- (4) System directory of the 16-bit Windows OS
No Win32 function is available to get the path of this directory, but the path can be searched. The directory name is SYSTEM.
- (5) Windows directory
The path of this directory can be obtained by the "GetWindowsDirectory" function.
- (6) Directory listed in the PATH environment variable

As described in "3. SOFTWARE CONFIGURATION," it is recommended to place the CHCUSB.dll in a folder that contains the application program (.EXE).

6.3.2 Loading the DLL dynamically

The application program can control the timing to load and close the DLL for initialization at the startup of the application program or when the application program does not use the functions of the CHCUSB DLL anymore.

The CHCUSB DLL is loaded, used, and closed in the following procedure:

- (1) Load CHCUSB.dll by the "LoadLibrary" function and get the module handle.

Example:

```
HANDLE hModule;
if ( (hModule = LoadLibrary( "CHCUSB.dll" )) == NULL ) {
    // LoadError
}
```

- (2) Get the pointer (address) of the exported DLL function (see 6.2) by the "GetProcAddress" function of windows API. (Add "APIENTRY" as the function type.)

Example:

```
typedef BOOL (APIENTRY *CHCUSB_OPEN_PROC)( WORD *rResult );
CHCUSB_OPEN_PROC chcusb_openFunc;
if ((chcusb_openFunc =
(CHCUSB_OPEN_PROC)GetProcAddress(hModule,"chcusb_open"))==NULL ) {
    // FunctionPointerGetError
}
```

- (3) Call the function by its pointer and use it.

Example:

```
WORD rResult;
if ( chcusb_openFunc( &rResult ) == FALSE ) {
    // CHCUSB DLLInitializeError
}
```

- (4) Discard the loaded DLL by the "FreeLibrary" function of the Windows API.

Example:

```
FreeLibrary( hModule );
(Execute the "chcusb_close" function before discarding the DLL.)
```


6.4 Image Data Formats Provided by the Application Program

The “chcusb_write” function of the CHCUSB DLL supports the following image data formats:

- RGB point sequential raw data format
- BGR point sequential raw data format

6.4.1 RGB point sequential raw data format

This data format represents one image by an array of level values (0 to 255) of color components R (red), G (green) and B (blue) that are arranged in that order. (See the table below.) The “chcusb_write” function sends a total of the image width (in dots) by the height of image (in lines) $\times$ 3 bytes.

Pixel 1			Pixel 2				Pixel N			
R11	G11	B11	R12	G12	B12		R1N	G1N	B1N	Line 1
R21	G21	B21	R22	G22	B22		R2N	G2N	B2N	Line 2
R31	G31	B31	R32	G32	B32		R3N	G3N	B3N	Line 3
⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮	⋮
RL1	GL1	BL1	RL2	GL2	BL2		RLN	GLN	BLN	Line L

To use this data format, specify the following value for the first argument of the “chcusb_imageformat” function.

Constant definition	Value	Meaning
kFORMAT_PIXEL_RGB	3	RGB point sequential raw data format

6.4.2 BGR point sequential raw data format

This data format represents one image by an array of level values (0 to 255) of color components B (blue), G (green) and R (red) that are arranged in that order. (See the table below.)

The “chcusb_write” function sends a total of the image width (in dots) by the height of image (in lines) $\times$ 3 bytes. The order of color components in this data format is reverse to that of color components of the RGB point sequential raw data format.

Pixel 1			Pixel 2				Pixel N			
B11	G11	R11	B12	G12	R12		B1N	G1N	R1N	Line 1
B21	G21	R21	B22	G22	R22		B2N	G2N	R2N	Line 2
B31	G31	R31	B32	G32	R32		B3N	G3N	R3N	Line 3
⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮	⋮
BL1	GL1	RL1	BL2	GL2	RL2		BLN	GLN	RLN	Line L

To use this data format, specify the following value for the first argument of the “chcusb_imageformat” function.

Constant definition	Value	Meaning
kFORMAT_PIXEL_BGR	4	BGR point sequential raw data format

6.5 List of Functions

The table below lists functions, their works, I/O parameters, and abilities to communicate with printers.

Function	Definition	Parameter		Communication with printer
		Input	Output	
chcusb_makeThread	Defined maximum thread number of Multi-Thread.	Maximum thread number	None	-
chcusb_open	Initializes the CHCUSB DLL and generates new thread.	None	Executing status	Enabled
chcusb_close	Closes the current thread, and closes the work memory and DLLs that the CHCUSB DLL uses.	None	None	-
chcusb_releaseThread	Closes all threads, and closes the work memory and DLLs that the CHCUSB DLL uses.	None	None	-
chcusb_listupPrinter	Lists printer IDs of printers that are connected to the computer.	None	Number of printers connected to the computer, printer ID number string	Enabled
chcusb_listupPrinterSN	Lists serial numbers of printers that are connected to the computer.	None	Number of printers connected to the computer, printer serial number string	Enabled
chcusb_selectPrinter	Selects a printer (using printer ID) that the computer uses.	Printer ID number	Executing status, printer ID number	Enabled
chcusb_selectPrinterSN	Selects a printer (using printer serial number) that the computer uses.	Printer serial number	Executing status, printer serial number	Enabled
chcusb_getPrinterInfo	Gets the firmware version of the selected printer, the total print count, and USB information.	Information tag number and information buffer	Obtained information	Enabled
chcusb_imageformat	Sets an image size and a transfer mode.	Transfer format, number of color components, image width, and image height	Executing status	-
chcusb_setmtf	Sets parameters for profile emphasis and smoothing.	3×3MTF parameter	None	-
chcusb_makeGamma	Creates a tone curve table.	Gamma value	Tone curve data	-
chcusb_setlcctable	Specifies a color matching table and an I/O tone curve.	Input profile name, output profile name, matching option, input tone curve, and output tone curve	Executing status	-
chcusb_copies	Sets the number of copies.	Number of copies	Executing status	-
chcusb_status	Checks the status of the currently-selected printer.	None	Executing status	Enabled
chcusb_statusAll	Checks the status of the specified printer.	None	Executing status	Enabled
chcusb_startpage	Starts page processing and sends an image header and print parameters.	None	Executing status	Enabled
chcusb_endpage	Ends page processing.	None	Executing status	-

Function	Work	Parameter		Communication with printer
		Input	Output	
chcusb_write	Transmits image data.	Image data buffer and data length	Executing status	Enabled
chcusb_setPrinterInfo	Sets parameters such as media type for the printer.	Parameter types and parameters	Executing status	Enabled
chcusb_getGamma	Gets an I/O tone curve table from a file.	File name	Tone curve data	-
chcusb_getMtf	Gets MTF parameters from a file.	File name	MTF parameter	-
chcusb_cancelCopies	Cancels the print count setting for the image which is being printed, or is to be printed next, and makes the printer ready when the printout of the current page ends.	None	Executing status	Enabled
chcusb_setPrinterToneCurve	Changes YMC tone curve inside the printer.	Tone curve data	Executing status	Enabled
chcusb_getPrinterToneCurve	Acquires YMC tone curve inside the printer.	None	Tone curve data	Enabled
chcusb_blinkLED	Flashes LED of selected printer.	None	Executing status	Enabled
chcusb_resetPrinter	Resets selected printer.	None	Executing status	Enabled
Chcusb_attchThreadCount	Checks the thread count of the used and maximum set.	None	Opened thread counts, Set the maximum thread count	-
chcusb_getColorSensorLevel	Acquires color sensor Level of the ribbon set to the printer.	None	Color sensor levels	Enabled

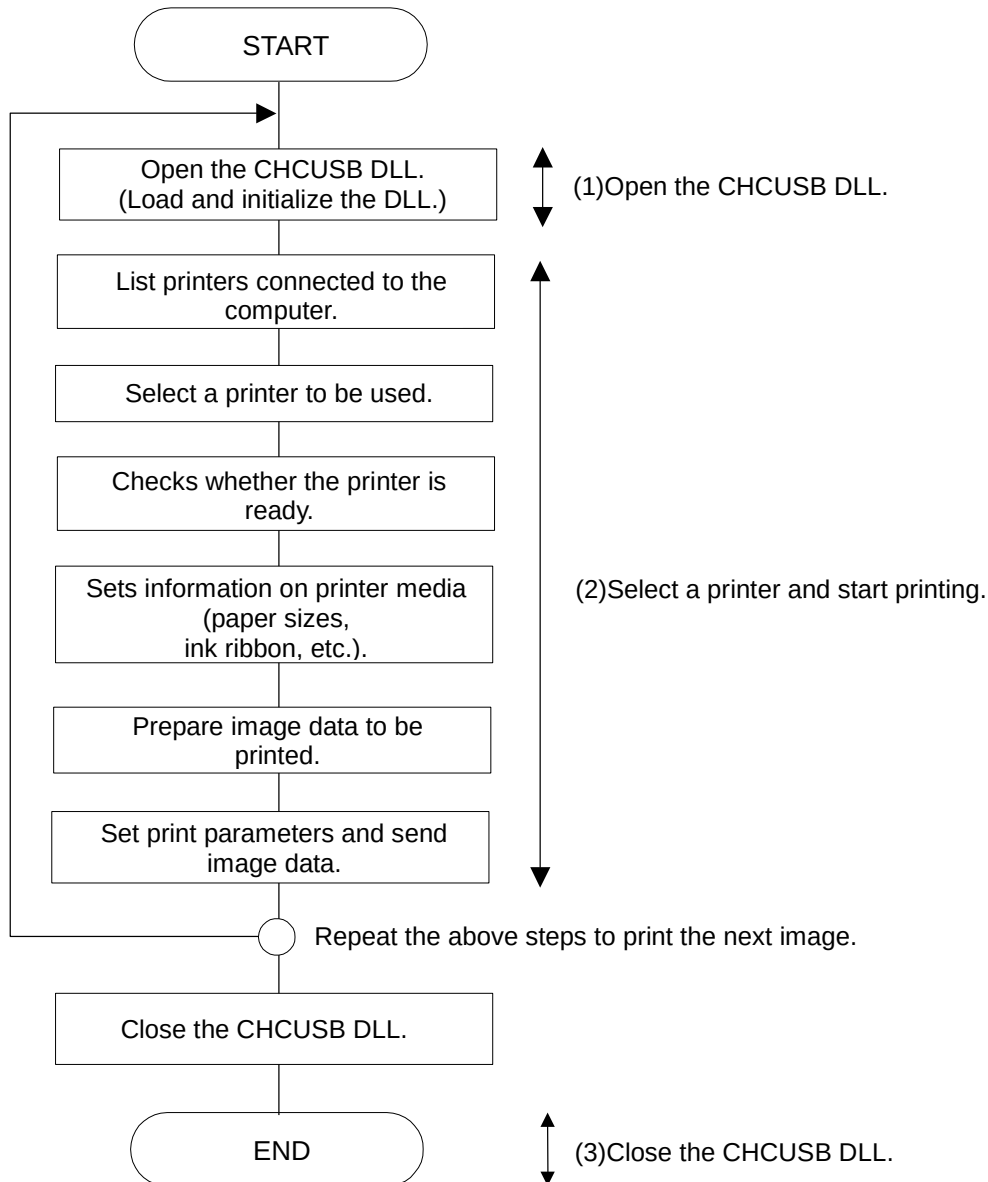
The functions capable of communicating with a printer (excluding the “chcusb_statusAll” function), communicate with a printer selected by the “chcusb_selectPrinter” or “chcusb_selectPrinterSN” function.

6.6 Process Flow

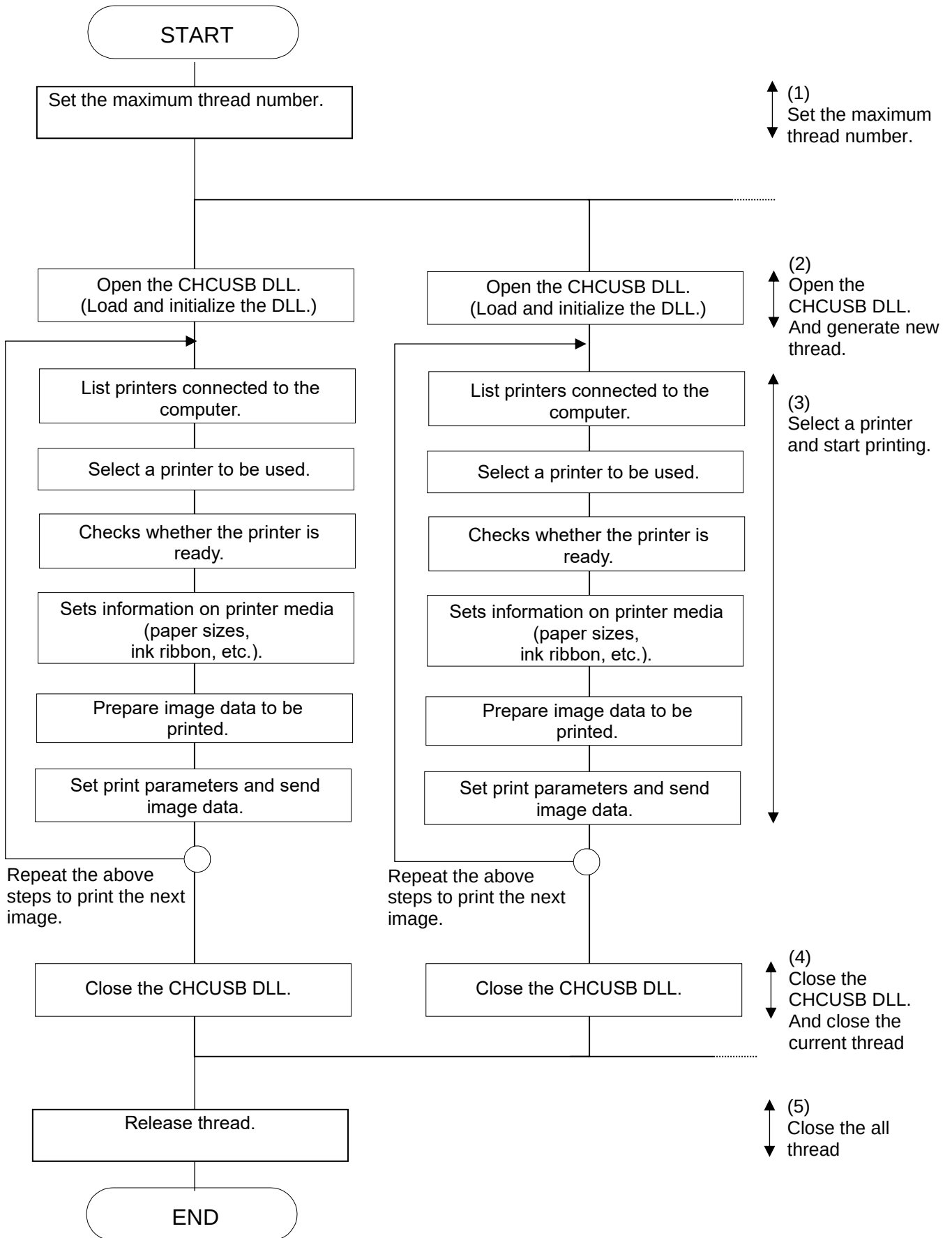
This section describes how the CHCUSB DLL is controlled to open, initialize, select a printer, start printing, and close the DLL.

6.6.1 Process flow

A. Single thread



B. Multi threads



6.6.2 Opening and closing the CHCUSB DLL

The following functions are called to open and close the CHCUSB DLL:

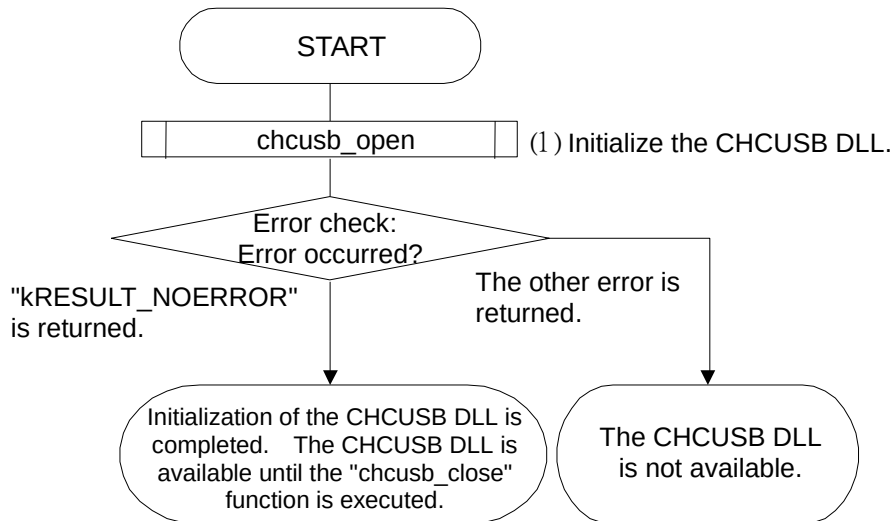
Function called to open the CHCUSB DLL: "chcusb_open" function

Function called to close the CHCUSB DLL: "chcusb_close" function

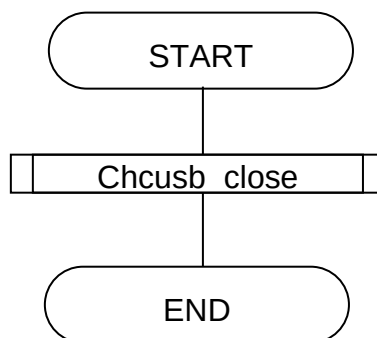
These functions are called in the following two ways:

- (1) Executing the "chcusb_open" function at the startup of the application and reading the "chcusb_close" function before closing the application
- (2) Executing the "chcusb_open" function when printing an image and reading the "chcusb_close" function at the end of printing.

Select one of the above two ways according to the system you want to install the CHCUSB DLL.



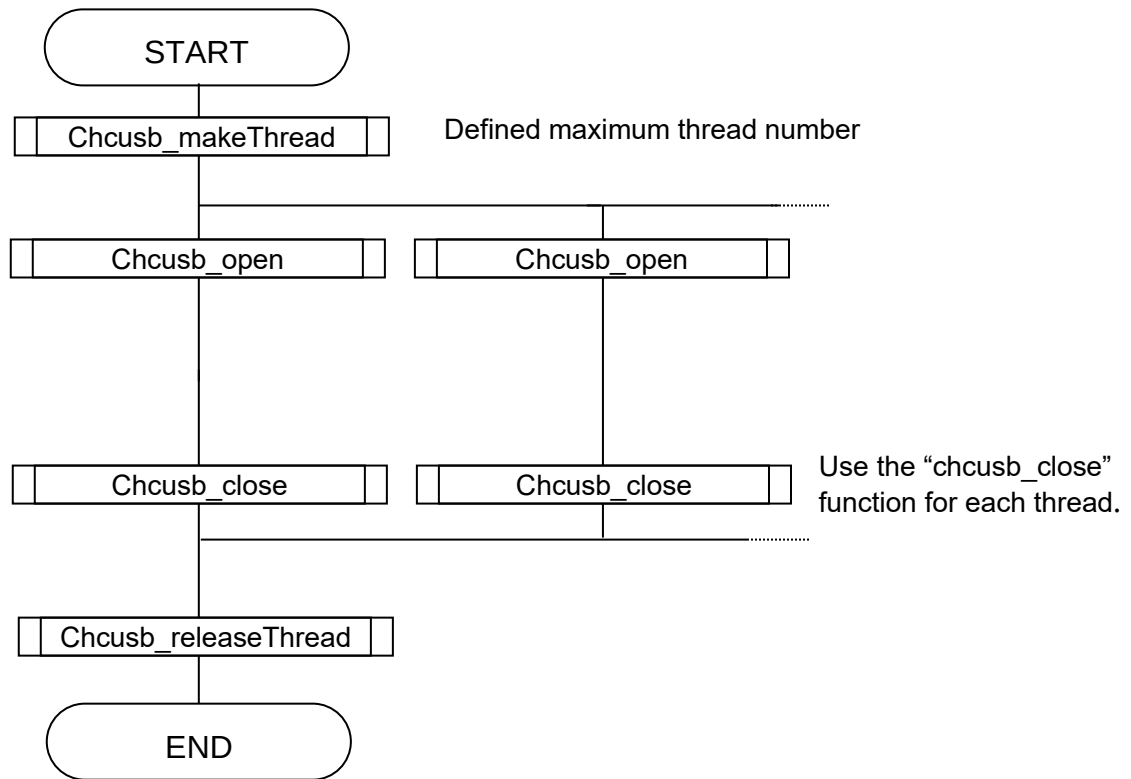
Starting to use the CHCUSB DLL



Ending the use of the CHCUSB DLL

Using multi-thread, it is the defined maximum thread number that uses the “chcusb_makeThread” function, before executing the “chcusb_open” function.

To close the Multi-thread, it uses the “chcusb_close” function for each thread.
After that, it uses the “chcusb_releaseThread” function.



Using multi-thread

6.6.3 Listing printers that are connected to the computer and selecting a printer

Call the “chcusb_open” function to initialize the CHCUSB DLL. Next, call the “chcusb_listupPrinter” function or “chcusb_listupPrinterSN” function to create a list of printers that are connected to the computer. The printer ID number list or printer serial number list created by the “chcusb_listupPrinter” function or “chcusb_listupPrinterSN” function is passed on, as it is to the “chcusb_statusAll” function to check the status for all printers.

After the printer ID number list or printer serial number list is created, call the “chcusb_selectPrinter” function or “chcusb_selectPrinterSN” function to select the printer to use from the list.

The “chcusb_listupPrinter” and “chcusb_listupPrinterSN” functions search for all devices connected via USB every time they are called upon, and the printer ID numbers or printer serial numbers are listed in the order that they are recognized.

For this reason, if a printer which existed at the time of the first call did not get recognized in the second call due to its power being turned off, it will not be included in the list.

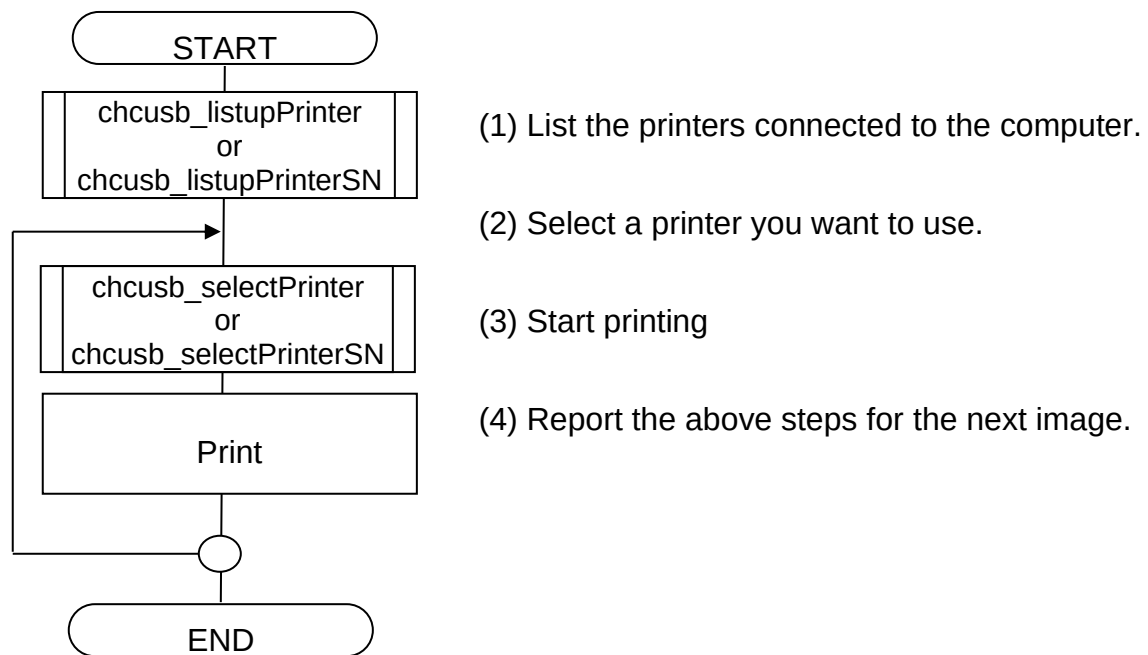
Printer ID number is BYTE type values of 0 through 127.

There are two ways of selecting a printer or printers to print to.

- Selecting one routine printer
- Selecting two or more printers which are ready but not busy.

6.6.3.1 Selecting one routine printer

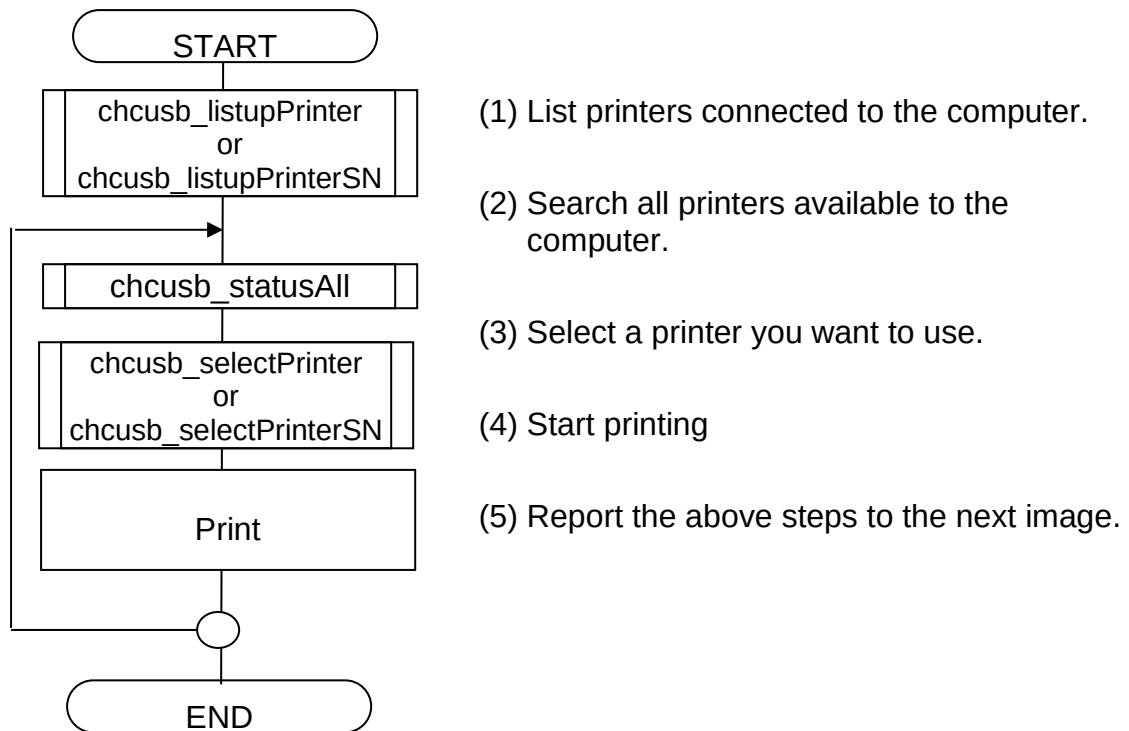
A flow to select one routine printer is shown below.



Selecting one routine printer to print to.

6.6.3.2 Selecting two or more printers which are ready but not busy

A flow to select two or more printers that are ready, but not busy is shown below.



Selecting two or more printers, which are ready but not busy.

It is also possible to select a printer by using the “chcusb_selectPrinter” function or “chcusb_selectPrinterSN” function (as described in 6.6.3.1), check whether the selected printer is ready by the “chcusb_status” function, and select the other printer by the “chcusb_selectPrinter” function or “chcusb_selectPrinterSN” function if the selected printer is not ready.

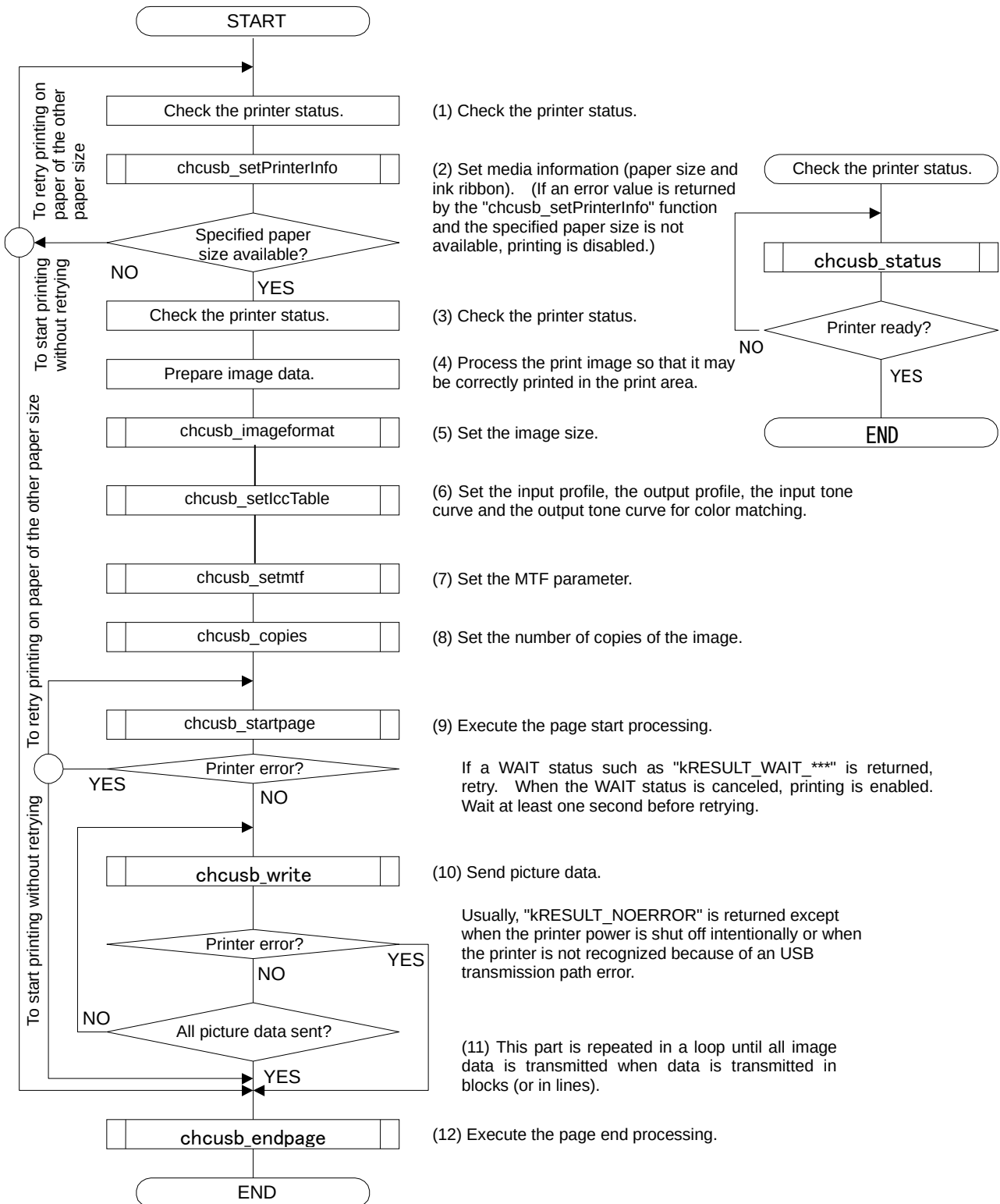
6.6.4 Printout sequence

After a printer is selected, the CHCUSB DLL enters a printout sequence, as shown below.

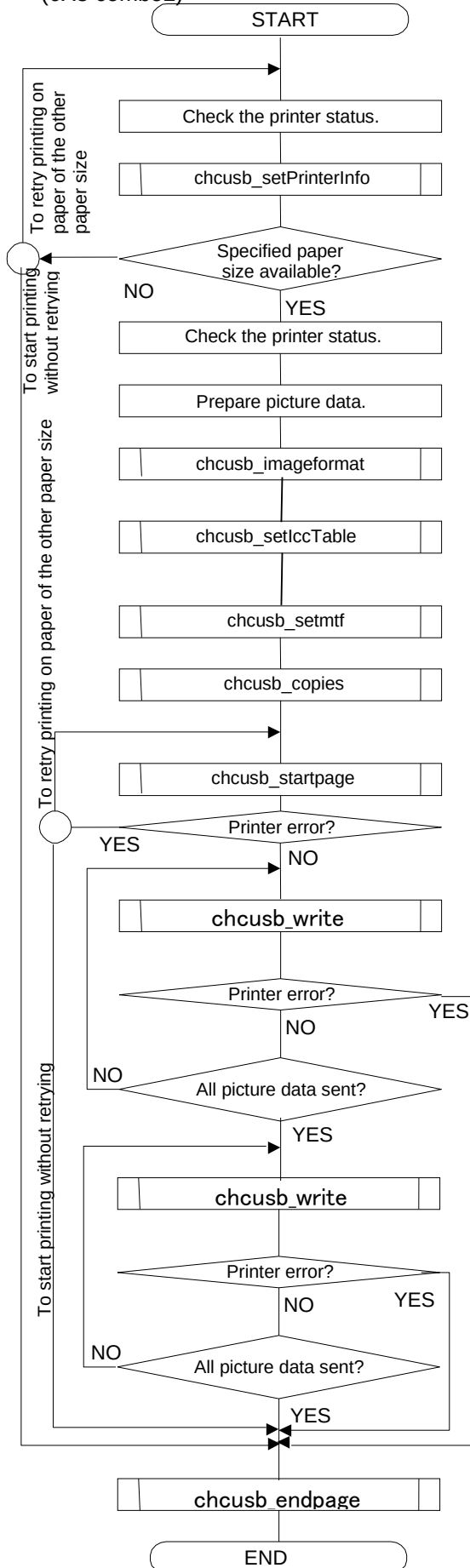
- (1) Call the "chcusb_status" function and make sure the selected printer is ready ("kRESULT_READY" is set in "rResult").
- (2) Set the printer media information (paper size and ink ribbon) by the "chcusb_setPrinterInfo" function. If the specified paper size with a "kPINFTAG_PAPER" tag identifier is not available to the selected printer, "kRESULT_UNKNOWNPAPER" (2301) is set in the returned executing status "rResult" and printing discontinues.
- (3) The application prepares the image data in a specified data format (see 6.4) according to the print area of the printer.
- (4) When parameters are specified in (2), the printer becomes busy until the parameters are set on the printer. Call the "chcusb_status" function and wait until the printer becomes ready.
- (5) Specify the size and the transfer format of the image to be printed by the "chcusb_imageformat" function. This image size must be equal to that specified by the "chcusb_setPrinterInfo" function.
- (6) Specify the input profile, the output profile, the input tone curve and the output tone curve for color matching by the "chcusb_setlccTable" function. When you want to use the parameters repeatedly to print, you can set them after executing the "chcusb_open" function. You need not specify them again because the setting is retained until the "chcusb_open" function is executed.
- (7) Set the profile emphasis/smoothing (MTF) parameters by the "chcusb_setmtf" function. When you want to use the parameters repeatedly to print out, you can set them after executing the "chcusb_open" function. You need not specify them again because the setting is retained until the "chcusb_open" function is executed.
- (8) Specify the number of copies of the image that you want to send to the printer.
- (9) Execute the "chcusb_startpage" function and send the image header and the print parameters.
- (10) Execute the "chcusb_write" function to send the image data. If using the MTF function, the color matching function, and the tone curve function after setting parameters for the "chcusb_setmtf" function and the "chcusb_setlccTable" function, the size of image data specified by the "chcusb_write" function must be a multiple of 4 times the image width. If not, the MTF, color matching, and tone curve functions will be disabled.
- (11) Step (10) above, is repeated until the whole one image data is sent.
- (12) Execute the "chcusb_endpage" function to end page processing.
- (13) Repeat (1) through (12) above for the same number of times the image is to be printed.

For printing an image, a flow to "printing out one image" is shown below.

Printing out one image
(4X6, 5X3.5, 6X8, 6X8 combo1, 6X8Border)



Printing out two images (6X8 combo2)



(1) Check the printer status.

(2) Set media information (paper size and ink ribbon). (If an error value is returned by the "chcusb_setPrinterInfo" function and the specified paper size is not available, printing is disabled.)

(3) Check the printer status.

(4) Process the print image so that it may be correctly printed in the print area.

(5) Set the image size.

(6) Set the input profile, the output profile, the input tone curve and the output tone curve for color matching.

(7) Set the MTF parameter.

(8) Set the number of copies of the image.

(9) Execute the page start processing.

If a WAIT status such as "kRESULT_WAIT_****" is returned, retry. When the WAIT status is canceled, printing is enabled. Wait at least one second before retrying.

(10) Send picture data 1.

Usually, "kRESULT_NOERROR" is returned except when the printer power is shut off intentionally or when the printer is not recognized because of an USB transmission path error.

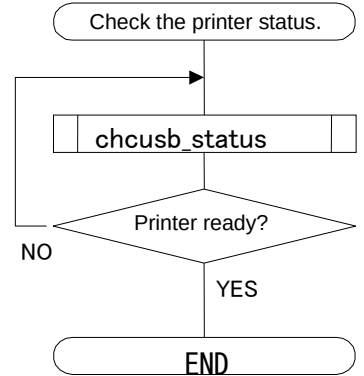
(11) This part is repeated in a loop until all picture data is transmitted when data is transmitted in blocks (or in lines).

(12) Send picture data 2.

Usually, "kRESULT_NOERROR" is returned except when the printer power is shut off intentionally or when the printer is not recognized because of an USB transmission path error.

(13) This part is repeated in a loop until all image data is transmitted when data is transmitted in blocks (or in lines).

(14) Execute the page end processing.



6.6.5 Job cancellation

Job cancellation can be done by canceling the print request using page the ID acquired by the "chcusb_startpage" function (however, it excludes the ongoing print (1 copy))

By executing the "chcusb_cancelCopies" function, a copy count is assigned to the printer to cancel the printing. However, the one print which is currently printing shall be continue the printing process.

Note:

This function is disabled while the printer is receiving data.

6.6.6 Print Numbers Reference Function

By assigning and executing a tag identifier kPINFTAG_MEMORY, by the "chcusb_getPrinterInfo" function, the page ID left in the memory bank and the number of prints left will be available for browsing.

Note:

- The number of prints left in the memory bank includes the print which is ongoing]
- When this function is used after executing the "chcusb_copies" function and the job is cancelled, the "chcusb_getPrinterInfo" function gets information about the number of prints left before the cancellation of the job, until the ongoing print process is finished.

6.6.7 Function to Change Tone Curve inside the Printer

Inside the printer is a tone curve table.

Use the "chcusb_setPrinterToneCurve" function to change the tone curve set up inside the printer.

Use the "chcusb_getPrinterToneCurve" function to browse the tone curve currently set up in the printer.

7. FUNCTION REFERENCE

Function name

chcusb_makeThread

Format

BOOL APIENTRY chcusb_makeThread(WORD maxCount);

Function

Defined maximum thread number for Multi-Thread.

Input

maxCount	Maximum thread number. The possible values are 1 to 100.
----------	---

Output

None

Remarks

Execute this function when using a multi-thread.

Function name
chcusb_open

Format
BOOL APIENTRY chcusb_open(WORD *rResult);

Function
Initializes the executing environment of the CHCUSB DLL and generates a new thread.

Input
rResult Pointer to a word type variable for storing the executing status

Output
Return value Returns "true" when the function ends normally or "false" if the other status.
rResult Executing status

The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("True" is returned.)
kRESULT_MEMFULLERR	1000	Memory Full
kRESULT_USBNOTFOUN D	1002	Cannot load SDKUsbCtrl.DLL.
kRESULT_ICCNOTFOUND	1003	Cannot load SDKColorMatch.DLL.
kRESULT_OPENPORT	1004	Cannot open the USB port.
KRESULT_THREAD_Max	1501	Cannot generate new thread.

Remarks
Execute the "chcusb_makeThread" function when using a multi-thread.

The "chcusb_open" function loads and initializes the following DLL files:
SDKColorMatch.DLL
SDKUsbCtrl.DLL

Function name
chcusb_close

Format
void APIENTRY chcusb_close(void);

Function
Closes the work memory and DLL files that the CHCUSB DLL used.

Input
None

Output
None

Remarks
Execute this function when all print jobs have been completed and the CHCUSB DLL is not used any more or to close the current thread.

Function name
chcusb_releaseThread

Format
BOOL APIENTRY chcusb_releaseThread(WORD* rResult);

Function
Closes the all threads, and closes the work memory and DLL files that the CHCUSB DLL used.

Input
rResult Pointer to the word type variable for storing the executing status

Output
rResult Executing status

The following states can be stored:

Status	Value	Description
KRESULT_READY	0	The printer is ready.
KRESULT_MUTEX_Used	1502	Other threads are executing.

Remarks
Execute this function when all print jobs have been completed and the CHCUSB DLL is not used any more, or to close the application program that installed the CHCUSB DLL.

Function name
chcusb_listupPrinter

Format
 WORD APIENTRY chcusb_listupPrinter(BYTE *rIdArray)

Function
 To search for all USB devices connected to the computer. It returns the number of printers available for use with CHCUSB DLL. At the same time, it creates a list of printer ID numbers of such printers.

Printer ID numbers are BYTE type values of 0 through 127, and they are numbers assigned in the order which CHCUSB DLL recognizes a printer, starting from 0, 1, 2..., when the "chcusb_listupPrinter" function is called.

Input
 rIdArray Pointer to 128-element byte type array to store printer ID number.

Output
 Return value Number of printers connected to the machine.

rIdArray Printer ID number list of the printers connected to the computer

rIdArray value is formatted as shown below.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	First printer ID number 0
1	BYTE	1	Second printer ID number 1
:	:	:	:
127	BYTE	1	128 th printer ID number 127

The rIdArray element description values for printers which are not connected will be FFh.

Note
 By assigning the printer ID number acquired by the "chcusb_selectPrinter" function, the printer you wish to use is selected.

In case a multiple number of printers are connected to the computer, the order that the CHCUSB DLL recognizes printers may differ due to the connection environment, therefore there is no guarantee which printer will be listed first.

Function name
chcusb_listupPrinterSN

Format
WORD APIENTRY chcusb_listupPrinterSN(SerialNumRec rSerialArray[])

Function
To search for all USB devices connected to the computer. It returns the number of printers available for use with CHCUSB DLL. At the same time, it creates a list of printer serial numbers of such printers.

Input
rSerialArray Pointer to an array of structs.

Output
Return value Number of printers connected to the machine.

rSerialArray Printer serial number list of the printers connected to the machine

rSerialArray value is formatted as shown below.

Offset	Type	Size (in bytes)	Description
0	SerialNumRec	8	First printer serial number
1	SerialNumRec	8	Second printer serial number
:	:	:	:
127	SerialNumRec	8	128 th printer serial number

The rSerialArray element description values for printers which are not connected will be FFh.

Note
By assigning the printer serial number acquired by chcusb_selectPrinterSN function, the printer you wish to use is selected.

In case a multiple number of printers are connected to the computer, the order that the CHCUSB DLL recognizes printers may differ due to the connection environment, therefore there is no guarantee which printer will be listed first.

Function name
chcusb_selectPrinter

Format
BYTE APIENTRY chcusb_selectPrinter(BYTE printerId, WORD *rResult);

Function
The printer with the specified printer ID number will be selected.

Input

printerId	Specify the printer ID number of the printer you want to select.
rResult	Pointer to a word type variable that stores the executing status.

Output

Return value	Returns selected printer ID number. In case FFh is assigned to the printerId, it returns the printer ID number of the printer currently selected.(in case no printers are selected, it returns FFh)
rResult	Executing status

Status that are to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Printer is selected.
kRESULT_NOTFOUND	1008	Printer with selected number is not found.
KRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Printer cannot be recognized.

Note
Printer ID numbers are acquired by creating a list with the "chcusb_listupPrinter" function.

Even when kRESULT_NOTCONNECT is returned for rResult, the printer with the specified printer ID number will be selected.

Function name
chcusb_selectPrinterSN

Format
SerialNumPtr APIENTRY chcusb_selectPrinterSN(SerialNumRec printerSN, WORD *rResult);

Function
The printer with the specified printer serial number will be selected to be the output.

Input
printerSN Specify the printer serial number of the printer you want to select.
rResult Pointer to a word type variable that stores the executing status.

Output
Return value Pointer to the serial number struct that contains the selected printer serial number. If FFh is assigned to the printerSN, it returns a pointer to the printer serial number of the currently selected printer. If no printers are selected, it returns FFh.
rResult Executing status

Status that are to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Printer is selected.
kRESULT_NOTFOUND	1008	Printer with selected number is not found.
KRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Printer cannot be recognized.

Note
Printer serial numbers are acquired by creating a list with the “chcusb_listupPrinterSN” function.

Even when kRESULT_NOTCONNECT is returned for rResult, the printer with the specified printer serial number will be selected.

Function name
chcusb_getPrinterInfo

Format
BOOL APIENTRY chcusb_getPrinterInfo(WORD tagNumber, void *rBuffer, DWORD *rLen);

Function
Gets the media information, USB information, the serial number and firmware version, the total print count, the error history, the memory bank information, and label information for the currently-selected printer.

Input
tagNumber Specify the tag number of information you want to get.

The tag numbers that can be specified are as follows:

Tag identifier	Value	Information to be obtained
KPINFTAG_PAPER	0	Information about paper sheets for the currently-selected printer
KPINFTAG_PAPER2	10	Paper information of the currently-selected printer.
KPINFTAG_USBINQ	2	USB inquiry of selected printer.
KPINFTAG_ENGID	3	Serial number and firmware version of the currently-selected printer
KPINFTAG_PRINTCNT	4	Total print count of the currently-selected printer
KPINFTAG_TABLEINFO	5	Print parameter software version of the currently-selected printer
KPINFTAG_SVCINFO	7	Detailed status at the occurrence of a Service-Call error (kRESULT_SVCCALL)
KPINFTAG_MEMORY	16	Number of prints to be printed for each memory bank.
KPINFTAG_LABELINFO	25	Label information of the currently-selected printer
KPINFTAG_ERRHISTORY	50	Error history information of the currently-selected printer
KPINFTAG_BUFFSW	60	4Buffers setting information of the currently-selected printer
KPINFTAG_WAITTIME	61	Wait time information of the currently-selected printer (For 4 Buffer mode)
KPINFTAG_DENSITY	62	Density information of the currently-selected printer
KPINFTAG_MEMORY2	17	Number of prints to be printed for each memory bank. (For 4 Buffer mode)
KPINFTAG_OFFSET	66	Offset information of the currently-selected printer
KPINFTAG_ENERGY_SAVE_MODE_ENABLED	101	Energy save mode enabling/disabling.
KPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME	104	Wait time [min] to enter energy save mode.

When the status values listed below are returned by the "chcusb_status" or "chcusb_statusAll" function, you can specify "KPINFTAG_SVCINFO" to get the detailed status. For other values, the content of the detailed status is indefinite.

The error condition which can obtain detailed status is as follows.

(New detailed status)

Status	Value
kRESULT_SVCCALL	2500
kRESULT_WAIT_WindingRibbonErr	2202
kRESULT_WAIT_OffLineErr	2204
kRESULT_WAIT_InitializeErr	2205
kRESULT_STATUS_CoverOpenErr	2401
kRESULT_STATUS_PaperCoverOpenErr	2402
kRESULT_STATUS_IncorrectRibbonErr	2403
kRESULT_STATUS_IncorrectPaperErr	2404
kRESULT_STATUS_RibbonEmptyErr	2405
kRESULT_STATUS_PaperEmptyErr	2408
kRESULT_STATUS_ReadyLoadingErr	2412
kRESULT_STATUS_TakeOutPaperErr	2413

(Old detailed status)

Status	Value
kRESULT_SVCCALL	2500
kRESULT_WAIT_WindingRibbonErr	2202
kRESULT_WAIT_OffLineErr	2204
kRESULT_WAIT_InitializeErr	2205
kRESULT_OPECALL_PaperEmptyErr	2305
kRESULT_OPECALL_InkEmptyErr	2306
kRESULT_OPECALL_PaperJamErr	2307
kRESULT_OPECALL_UpperCoverOpenErr	2309

The firmware version of the printer whose "New detailed status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version 237 or a newer version.
6800 Type 2	Version 337 or a newer version.
6800 Type 3	Version 437 or a newer version.
6850	Version 657 or a newer version.

The firmware version of the printer whose "Old detailed status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version of 236 or older version.
6800 Type 2	Version of 336 or older version.
6800 Type 3	Version of 436 or older version.
6850	Version of 656 or older version.

rBuffer Pointer to memory that stores information to be obtained

The size of memory required for each tag is as follows:

Tag identifier	Memory size required (in bytes)
kPINFTAG_PAPER	121
kPINFTAG_PAPER2	123
kPINFTAG_USBINQ	48
kPINFTAG_ENGID	24
kPINFTAG_PRINTCNT	20
kPINFTAG_TABLEINFO	2
kPINFTAG_SVCINFO	6
kPINFTAG_MEMORY	16
kPINFTAG_LABELINFO	24
kPINFTAG_ERRHISTORY	61
kPINFTAG_BUFFSW	50
kPINFTAG_WAITTIME	50
kPINFTAG_DENSITY	50
kPINFTAG_MEMORY2	32
kPINFTAG_OFFSET	50
kPINFTAG_ENERGY_SAVE _MODE_ENABLED	1
kPINFTAG_ENERGY_SAVE _MODE_WAIT_TIME	1

When "NULL" is specified for "rBuffer," this function sets the size of memory (in bytes) required to get information in "rLen" without getting the information.

rLen Pointer to the double-word type variable that stores the size of data to be read (in bytes)

Output

Return value Returns "TRUE" when the function ends normally or "FALSE" when an undefined tag identifier is specified or when the communication with the printer is disabled.

rLen Stores the length (in bytes) of information obtained by this function.
When "NULL" is specified for "rBuffer," this function sets the size of memory (in bytes) required to get information in "rLen" without getting the information.
Stores 0 when an undefined tag identifier is specified.

rBuffer Information obtained
Each tag identifier is formatted as shown below.

Tag identifier: kPINFTAG_PAPER

It stores information of printable size, based on media currently set in the printer. If there are more than 2 printable sizes, then it will be stored at Media Information 2, 3, etc.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Number of Media Information Number of media information is stored in binary value.
1	-	10	Media Information 1
11	-	10	Media Information 2
21	-	10	Media Information 3
31	-	10	Media Information 4
41	-	10	Media Information 5
51	-	10	Media Information 6
61	-	10	Media Information 7
71	-	10	Media Information 8
81	-	10	Media Information 9
91	-	10	Media Information 10
101	-	10	Media Information 11
111	-	10	Media Information 12

The following is the media information format.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Print area information code 00h ... info 01h ... info 06h ... info 07h ... info
1	WORD	2	Width of the print area (in dots)
3	WORD	2	Height of the print area (in lines)
5	BYTE	1	Medium type 00h ... Unknown 01h ... Paper
6	BYTE	1	Print type information code 00h ... normal print 01h ... Combo – 1 image print 02h ... Combo – 2 image print 03h ... print with border 04h ... Border Combo – 1 image print 05h ... Border Combo – 2 image print 20h ... Combo – 2 image print(2x6-2up) 21h ... Combo – 1 image print(2x6-1up) 22h ... Combo – 2 image print(3x6-2up) 23h ... Combo – 1 image print(3x6-1up)
7	BYTE	1	00h: Ribbon Position: Full Patch 01h: Ribbon Position: Half Patch
8	BYTE	2	Reserved

Note: When the above print area returns "0000h," the selected printer does not support getting the print area. When "FFFFh" is returned, it is assumed that the paper is unknown.

Note: Media information is different from the media loaded in the printer.
The information is transmitted from the printer as follows.

Media (Loaded in printer)	Media Information No.	Print area information code	Width of the print area (in dots)	Height of the print area (in lines)	Media type	Print type information code	Note
5X3.5	1	01h	1548 (060Ch)	1088 (0440h)	01h	00h	5X3.5print
6X4	1	00h	1844 (0734h)	1240 (04D8h)	01h	00h	6X4print
5x7	1	00h	1548 (060Ch)	2140 (085Ch)	01h	00h	5X7print
6X8	1	06h	1844 (0734h)	2434 (0982h)	01h	00h	6X8print
	2	06h	1844 (0734h)	1240 (04D8h)	01h	01h	6X8combo1 print (6X4 1print)
	3	06h	1844 (0734h)	1240 (04D8h)	01h	02h	6X8combo2 print (6X4 2prints)
	4	06h	1500 (05DCh)	2100 (0834h)	01h	03h	5X7 with border print
	5	06h	1844 (0734h)	1240 (04D8h)	01h	04h	6X8combo1 print (5X3.5 1print)
	6	06h	1844 (0734h)	1240 (04D8h)	01h	05h	6X8combo1 print (5X3.5 1print)
	7	06h	1844 (0734h)	2222 (08AEh)	01h	00h	152x185mm print
	8	06h	1844 (0734h)	1836 (072Ch)	01h	00h	6X6print
	9	06h	1844 (0734h)	1282 (0502h)	01h	20h	6x8combo2 print (2x6 2 prints)
	10	06h	1844 (0734h)	636 (027Ch)	01h	21h	6x8combo1 print (2x6 1 print)
	11	06h	1844 (0734h)	1882 (075Ah)	01h	22h	6x8combo2 print (3x6 2 prints)
	12	06h	1844 (0734h)	936 (03A8h)	01h	23h	6x8combo1 print (3x6 1 print)

When 6x8 Media type is selected
In case of Ribbon Position: Full Patch

Media Information No.	Print area information code	Width of the print area (in dots)	Height of the print area (in lines)	Media type	Print type information code	Ribbon Position	Note
1	06h	1844 (0734h)	2434 (0982h)	01h	00h	0	6X8print
2	06h	1844 (0734h)	1240 (04D8h)	01h	01h	0	6X8combo1 print (6X4 1print)
3	06h	1844 (0734h)	1240 (04D8h)	01h	02h	0	6X8combo2 print (6X4 2prints)
4	06h	1500 (05DCh)	2100 (0834h)	01h	03h	0	5X7 with border print
5	06h	1844 (0734h)	1240 (04D8h)	01h	04h	0	6X8combo1 print (5X3.5 border 1print)
6	06h	1844 (0734h)	1240 (04D8h)	01h	05h	0	6X8combo2 print (5X3.5 border 2prints)
7	06h	1844 (0734h)	2222 (08AEh)	01h	06h	0	152x185mm print
8	06h	1844 (0734h)	1836 (072Ch)	01h	00h	0	6X6print
9	06h	1844 (0734h)	1282 (0502h)	01h	20h	0	6x8 Combo2 (2x6-2up)
10	06h	1844 (0734h)	636 (027Ch)	01h	21h	0	6x8 Combo1 (2x6-1up)
11	06h	1844 (0734h)	1882 (075Ah)	01h	22h	0	6x8 Combo2 (3x6-2up)
12	06h	1844 (0734h)	936 (03A8h)	01h	23h	0	6x8 Combo1 (3x6-1up)

In case of Ribbon Position: Half Patch

Media Information No.	Print area information code	Width of the print area (in dots)	Height of the print area (in lines)	Media type	Print type information code	Ribbon Position	Note
1	06h	1844 (0734h)	2434 (0982h)	01h	00h	0	6X8print
2	06h	1844 (0734h)	1240 (04D8h)	01h	01h	0	6X8combo1 print (6X4 1print)
3	06h	1844 (0734h)	1240 (04D8h)	01h	02h	1	6X8combo2 print (6X4 2prints)
4	06h	1500 (05DCh)	2100 (0834h)	01h	03h	0	5X7 with border print
5	06h	1844 (0734h)	1240 (04D8h)	01h	04h	0	6X8combo1 print (5X3.5 border 1print)
6	06h	1844 (0734h)	1240 (04D8h)	01h	05h	1	6X8combo2 print (5X3.5 border 2prints)
7	06h	1844 (0734h)	2222 (08AEh)	01h	06h	0	152x189mm print
8	06	1844 (0734h)	1836 (072Ch)	01h	00h	0	6X6print
9	06h	1844 (0734h)	1282 (0502h)	01h	20h	1	6x8 Combo2 (2x6-2up)
10	06h	1844 (0734h)	636 (027Ch)	01h	21h	0	6x8 Combo1 (2x6-1up)
11	06h	1844 (0734h)	1882 (075Ah)	01h	22h	1	6x8 Combo2 (3x6-2up)
12	06h	1844 (0734h)	936 (03A8h)	01h	23h	0	6x8 Combo1 (3x6-1up)

Tag identifier: kPINFTAG_ERRHISTORY

The error history information is stored in the printer. The last 10 error occurrences are stored in the printer, with the most recent occurrence listed first.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Command receive respond code
1	-	6	Error origin information ①
7	-	6	Error origin information ②
13	-	6	Error origin information ③
19	-	6	Error origin information ④
25	-	6	Error origin information ⑤
31	-	6	Error origin information ⑥
37	-	6	Error origin information ⑦
43	-	6	Error origin information ⑧
49	-	6	Error origin information ⑨
55	-	6	Error origin information ⑩

Command receive respond code (offset 0)

Code for checking if command from host is received successfully.

Value	Description
01h	OK...Received successfully
02h	NG...Cannot receive successfully

The following is format of Error information ①-⑩

Offset	Type	Size (in bytes)	Description
X*6+0	BYTE	1	Error kind information
X*6+1	BYTE	1	Error detailed cord
X*6+2	BYTE	1	The product arithmetic print number of sheets at the time of error occurrence
X*6+3	BYTE	1	
X*6+4	BYTE	1	
X*6+5	BYTE	1	

(* X corresponds to ①-⑩)

Tag identifier: kPINFTAG_PAPER2

It stores the print size information that can be printed using the media that are set in the printer. When multiple usable print sizes are present, the corresponding information is printed as media information (2) or (3).

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Code of ribbon set in Printer. 00h ... Ribbon is not loaded. 0Bh ... 6x8" 6SR 0Dh ... 6x8" 6PR 0Eh ... 5x7" 5PR 11h ... 6x8" 6TR Reserve1 14h ... 5x7" 5TR Reserve 1 17h ... 6x8" 6TR Reserve2 19h ... 5x7" 5TR Reserve 2 28h ... 6x8" 6TR Reserve3 2Bh ... 5x7" 5TR Reserve 3 2Ch ... 6x8" 6TR2 2Eh ... 5x7" 5TR2
1	BYTE	1	Reserved.
2	BYTE	1	Number of sets of media information The current number of sets of media information is stored as a binary-coded numeric value.
3	-	10	Media information (1)
13	-	10	Media information (2)
23	-	10	Media information (3)
33	-	10	Media information (4)
43	-	10	Media information (5)
53	-	10	Media information (6)
63	-	10	Media information (7)
73	-	10	Media information (8)
83	-	10	Media information (9)
93	-	10	Media information (10)
103	-	10	Media information (11)
113	-	10	Media information (12)

The format of media information is shown below.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Print area information code 00h ... 4x6 01h ... 5x3.5 06h ... 6x8 07h ... 5x7
1	WORD	2	Width of the print area (in dots)
3	WORD	2	Height of the print area (in lines)
5	BYTE	1	Medium type 00h ... Unknown 01h ... Paper
6	BYTE	1	Print type information code 00h ... normal print 01h ... Combo 1 image print 02h ... Combo 2 image print 03h ... print with border 04h ... Border Combo 1 image print 05h ... Border Combo 2 image print 20h ... Combo – 2 image print(2x6-2up) 21h ... Combo – 1 image print(2x6-1up) 22h ... Combo – 2 image print(3x6-2up) 23h ... Combo – 1 image print(3x6-1up)
7	BYTE	1	Ribbon Position 00h ... Full Patch 01h ... Half Patch
8	WORD	2	Reserved

Note: When the above print area returns "0000h," the selected printer does not support getting the print area. When "FFFFh" is returned, it is assumed that the paper is unknown.

The version which the printer of each model supports this function is as follows;

Model	Supported version(main control program version)
6800 Type 1	230
6800 Type 2	330
6800 Type 3	430
6850	650

Tag identifier: kPINFTAG_USBINQ

Offset	Type	Size (in bytes)	Description
0	WORD	2	Vendor ID
2	WORD	2	Product ID
4	BYTE	42	String obtained by string descriptor (40 bytes) plus termination code \0 (00h) x 2.
46-47	BYTE	2	Reserved

Tag identifier: kPINFTAG_ENGID

Firmware version is returned.

Offset	Type	Size (in bytes)	Description
0	WORD	2	Engine firmware (main boot program) version
2	WORD	2	Engine firmware (main control program) version
4	WORD	2	Engine firmware (input boot program) version
6	WORD	2	Engine firmware (input control program) version
8	BYTE	8	Printer serial number

Tag identifier: kPINFTAG_PRINTCNT

Offset	Type	Size (in bytes)	Description
0	DWORD	4	Total print count 1 (permanent)
4	DWORD	4	Total print count 2 (maintenance)
8	DWORD	4	Print count after media replacement
12	DWORD	4	Cutter count
16	DWORD	4	Remaining quantity of media

Tag identifier: kPINFTAG_MEMORY

Offset	Type	Size (in bytes)	Description
0	WORD	2	Page ID in memory bank 1
2	WORD	2	Page ID in memory bank 2
4	WORD	2	Number of prints left in memory bank 1
6	WORD	2	Number of completed prints in memory bank 1
8	WORD	2	Number of specified print count in memory bank 1
10	WORD	2	Number of prints left in memory bank 2
12	WORD	2	Number of completed prints in memory bank 2
14	WORD	2	Number of specified print count in memory bank 2

Note: If chcusb_cancelCopies is executed, note that the print number left, number of completed prints, and number of specified print count will all be set to 0.

Tag identifier: kPINFTAG_TABLEINFO

Version of software is returned.

Offset	Type	Size (in bytes)	Description
0	WORD	2	Print parameter software version

Note : In case of the printer which does not use Print parameter software, this information is stored 0000h.

Tag identifier: kPINFTAG_LABELINFO

Acquires label number, label names which were nominally set up.

(at factory: label number=0 label name=filled with blanks [0x20])

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Label number 0 ~ 255 binary
1~23	BYTE	23	Label name character string

Tag identifier: kPINFTAG_BUFFSW

Offset	Type	Size (in bytes)	Description
0	BYTE	1	4Buffers information 1:ON 0:OFF
1-49	BYTE	1	Reserved

Tag identifier: kPINFTAG_WAITTIME

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Wait time ranges from 0 to 100
1-49	BYTE	1	Reserved

Tag identifier: kPINFTAG_DENSITY

6850 printer model

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Density YMC ranges from 0 to 200
1	BYTE	1	Density LOW ranges from 0 to 250
2	BYTE	1	Density LAMI ranges from 0 to 221
3-49	BYTE	1	Reserved

6800 printer model

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Density YMC ranges from 0 to 200
1	BYTE	1	Density LOW ranges from 0 to 180
2	BYTE	1	Density LAMI ranges from 0 to 221
3-49	BYTE	1	Reserved

Tag identifier: kPINFTAG_OFFSET

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Side offset
1	BYTE	1	Top offset
2	BYTE	1	Sensor offset
3-49	BYTE	1	Reserved

The version which the printer of each model supports this function is as follows;

Model	Supported version(main control program version)
6800 Type 1	Version 196 or a later version.
6800 Type 2	Version 309 or a later version.
6800 Type 3	Version 409 or a later version.
6850	Version 614 or a later version.

Tag identifier: kPINFTAG_MEMORY2

Offset	Type	Size (in bytes)	Description
0	WORD	2	Page ID in memory bank 1
2	WORD	2	Number of prints left in memory bank 1
4	WORD	2	Number of completed prints in memory bank 1
6	WORD	2	Number of specified print count in memory bank 1
8	WORD	2	Page ID in memory bank 2
10	WORD	2	Number of prints left in memory bank 2
12	WORD	2	Number of completed prints in memory bank 2
14	WORD	2	Number of specified print count in memory bank 2
16	WORD	2	Page ID in memory bank 3
18	WORD	2	Number of prints left in memory bank 3
20	WORD	2	Number of completed prints in memory bank 3
22	WORD	2	Number of specified print count in memory bank 3
24	WORD	2	Page ID in memory bank 4
26	WORD	2	Number of prints left in memory bank 4
28	WORD	2	Number of completed prints in memory bank 4
30	WORD	2	Number of specified print count in memory bank 4

Note1: In case chcusb_cancelCopies is executed, note that the print number left, number of completed prints, number of specified print count, will all be set as 0.

Note2: If this function is called when the printer is 2 buffer mode, Offset16-31 will be set to 0.

Tag identifier: kPINFTAG_SVCINFO

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Status 1
1	DWORD	4	Status 2
5	BYTE	1	Detailed status

Status values and descriptions are listed below.

Status 1

Value	Description
00000001h	Ready
00000002h	Error
00000003h	Wait

Status 2 (when "Status 1" is "Ready")

Value	Description
00000000h	Ready

Status 2 (when "Status 1" is "Wait")

Value	Description
00000000h	Initial
00000001h	Rewinding the ribbon
00000002h	Thermal Protect
00000003h	Operation Block Busy

Status 2 (when "Status 1" is "Error")

(When an error occurs, the related error bit of Status 2 (32-bit flag) is turned on (= "1"). A bit value of "0" indicates that the related error has not occurred.)

Bit	Error status	Sense key detected by "Request Sens"
31	Control circuit error	Service-Call.(04 8d 04)
30	Mechanism control error	Service-Call.(04 8d 04)
29	--	--
28		
27		
26		
25		
24		
23		
22		
21		
20		
19		
18		
17		
16		
15		
14		
13	Sensor error	Service-Call.(04 8d 04)
12	Cover Open	Logical Unit Not Ready. Cover Open.(04 85 02)
11	--	--
10	Improper paper guide setting	Logical Unit Not Ready. Paper Guide Error.(04 93 02)
9	Thermal sensor error	
8	Paper jam	Logical Unit Not Ready. Paper Jam.(04 83 02)
7	--	--
6	No paper	Logical Unit Not Ready. Paper Out.(04 81 02)
5	--	--
4	No ink ribbon	Logical Unit Not Ready. Ink Ribbon Out.(04 82 02)
3	--	--
2		
1		
0	Receive data error	Logical Unit Not Ready. Data Error.(04 92 02)

Detailed status (when "Status 1" is "Error")

Status 2 error	Description
Sensor error	00h: No error 01h to FFh: Error number
Thermal sensor error	
Paper jam error	
Control circuit error	
Mechanism control error	

Error value Detailed Status

Status 2 error	Description	Value	Priority
Sensor error	Front End of Paper sensor	1	High
	Print Position sensor	2	
	No-Paper sensor	3	
	Paper Size sensor	4	
	Seal Paper Print Position sensor	5	
	Front End of Postal Card sensor	6	
	Ribbon sensor 1	7	Low
	Ribbon sensor 2	8	
	Ribbon sensor 3	9	
	Ribbon sensor 4	10	
	Takeup Motor Current sensor	11	
	Takeup Ribbon Diameter sensor	12	
Thermal sensor error	Ambient Temperature Thermistor 1 (Temperature Outside Media) High	1	High
	Ambient Temperature Thermistor 1 (Temperature Outside Media) Low	2	
	Ambient Temperature Thermistor 2 (Temperature Outside Printer) High	3	
	Ambient Temperature Thermistor 2 (Temperature Outside Printer) Low	4	Low
	Head Temperature Thermistor High	5	
	Head Temperature Thermistor Low	6	
Paper jam	Paper jam 1	1	This error will not occur together with the other error.
	Paper jam 2	2	
	Paper jam 3	3	
	Paper jam 4	4	
	Paper jam 5	5	
	Paper jam 6	6	
	Paper jam 7	7	
	Paper jam 8	8	
	Paper jam 9	9	
	Paper jam 10	10	
Control circuit error	Bus communication error (Main to Head)	1	High
	Bus communication error (Main to SCSI)	2	
	Serial Bus communication error (Main to Head)	3	
	Serial Bus communication error (Main to SCSI)	4	
	Flash ROM error 1	5	
	Flash ROM error 2	6	
	Flash ROM error 3	7	
	EEPROM error	8	
	DMA1 error	9	
	DMA2 error	10	
	DRAM memory error	11	Low
Mechanism control error	Head drive error	1	High
	Main pinch roll drive error	2	
	Feed pinch roll drive error	3	Low
	Cutter drive error	4	
	Paper feed mechanism error	5	

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_ENABLED

Offset	Type	Size (in bytes)	Description
1	BYTE	1	Energy save mode error enabling/disabling. - Disable is 0, Enable is 1.

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_ENABLED

Offset	Type	Size (in bytes)	Description
1	BYTE	1	Wait time [min] to enter energy save mode.

Remarks

(1) When CHCUSB DLL is inputted kPINFTAG_MEMORY2 to tagNumber in case of the printer does not support this function, CHCUSB DLL returns "FALSE" by return value.

The version which the printer of each model supports this function is as follows;

Model	Supported version(main control program version)
6800 Type 1	Version 193 or a later version.
6800 Type 2	Version 306 or a later version.
6800 Type 3	Version 406 or a later version.
6850	Version 609 or a later version.

Function name

chcusb_setPrinterInfo

Format

```
BOOL WINAPIENTRY chcusb_setPrinterInfo( WORD tagName, void *rBuffer, DWORD *rLen,
WORD *rResult );
```

Function

Sets media information of the currently-selected printer.

Input

tagNumber

Specify the tag number of information you want to specify.
The following tag numbers are available.

Tag identifier	Value	Information to be set
kPINFTAG_PAPER	0	Paper type information
kPINFTAG_POLISH	20	Laminate Print setting Information
kPINFTAG_LABELINFO	25	Printer label information
kPINFTAG_BUFFSW	60	Printer 4buffers information
kPINFTAG_WAITTIME	61	Printer wait time information
kPINFTAG_DENSITY_YMC	63	Printer density YMC information
kPINFTAG_DENSITY_LOW	64	Printer density LOW information
kPINFTAG_DENSITY_LAMI	65	Printer density LAMI information
kPINFTAG_SIDEOFFSET	67	Printer side offset information
kPINFTAG_TOPOFFSET	68	Printer top offset information
kPINFTAG_SENOROFFSET	69	Printer sensor offset information
kPINFTAG_ENERGY_SAVE_MODE_ENABLED	101	Energy save mode enabling/disabling.
kPINFTAG_ENERGY_SAVE_MODE_AWAKE	102	Wake printer up from energy save mode.
kPINFTAG_ENERGY_SAVE_MODE_SLEEP	103	Put printer into energy save mode.
kPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME	104	Wait time to enter energy save mode.(x10s)

rBuffer

Pointer to memory that stores information to be set

The data sizes of each tag to be specified by rLen are as follows.

Tag identifier	Size (in bytes)
kPINFTAG_PAPER	10
kPINFTAG_POLISH	2
kPINFTAG_LABELINFO	24
kPINFTAG_BUFFSW	2
kPINFTAG_WAITTIME	2
kPINFTAG_DENSITY_YMC	2
kPINFTAG_DENSITY_LOW	2
kPINFTAG_DENSITY_LAMI	2
kPINFTAG_SIDEOFFSET	2
kPINFTAG_TOPOFFSET	2
kPINFTAG_SENSOROFFSE T	2
kPINFTAG_ENERGY_SAVE_MODE_ENABLED	1
kPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME	1

Each tag is formatted as shown below.

Tag identifier: kPINFTAG PAPER

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Medium size information code
1	WORD	2	Width of the print area (in dots)
3	WORD	2	Height of the print area (in lines)
5	BYTE	1	Medium type
6	BYTE	1	Print type
7	BYTE	3	Reserved

Print area information code is defined to as follows:

Print area information code

Code	Description
00h	4X6
01h	5X3.5
06h	6X8
07h	5X7

The print area available is as follows:

(The actual paper sizes available are dependent upon printer model. See the operation manual or specifications of the printer.)

Paper size	Print area width (in dots)	Print area length (in lines)
4X6	1844 (0734h)	1240 (04D8h)
5X3.5	1548 (060Ch)	1088 (0440h)
6X8	1844 (0734h)	2434 (0982h)
6X8 Combo2 (4X6-2up)	1844 (0734h)	1240 (04D8h)
6X8 Combo1 (4X6-1up)	1844 (0734h)	1240 (04D8h)
6X8Border (5X7with Border)	1500 (05DCh)	2100 (0834h)
5X7	1548 (060Ch)	2140 (085Ch)
6X8 Combo2 (5X3.5-2up with Border)	1500 (05DCh)	1050 (041Ah)
6X8 Combo1 (5X3.5-1up with Border)	1500 (05DCh)	1050 (041Ah)
152x185mm	1844 (0734h)	2222 (08AEh)
6X6	1844 (0734h)	1836 (072Ch)
6x8 Combo2 (2x6-2up)	1844 (0734h)	1282 (0502h)
6x8 Combo1 (2x6-1up)	1844 (0734h)	636 (027Ch)
6x8 Combo2 (3x6-2up)	1844 (0734h)	1882 (075Ah)
6x8 Combo1 (3x6-1up)	1844 (0734h)	936 (03A8h)

Input (Continued)

Media type codes are as follows:

Media type

Code	Description
01h	Paper

Codes for print types are to be as follows:

Print type

Code	Description
00h	Normal print
01h	Combo 1 image print
02h	Combo 2 image print
03h	Print with border
04h	Border Combo 1 image print
05h	Border Combo 2 image print
20h	Combo 2 image print (2x6 2up)
21h	Combo 1 image print (2x6 1up)
22h	Combo 2 image print (3x6 2up)
23h	Combo 1 image print (3x6 1up)

Note: Setting the codes 01h ~ 05h, 20h ~ 23h is possible only when the media size information is 06h (6x8size). Note that in case of other paper sizes, an error may occur.

Tag identifier: kPINFTAG_POLISH

Laminate Print setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	0000h Print without lamination 0001h Print with lamination

Tag identifier: kPINFTAG_LABELINFO

Sets nominal label number and label name.

The set label number & label names are kept even when printer power is turned OFF.

(at factory: label number=0 label name=filled with blanks [0x20])

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Label number 0~255 binary
1~23	BYTE	23	Label name character string

Tag identifier: kPINFTAG_BUFFSW

4buffers setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	0000h 4buffers setting OFF 0001h 4buffers setting ON

Tag identifier: kPINFTAG_WAITTIME

Wait time setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Wait time ranges from 0 to 100

Tag identifier: kPINFTAG_DENSITY_YMC

Density YMC setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Density YMC ranges from 0 to 200

Tag identifier: kPINFTAG_DENSITY_LOW
Density LOW setting Information

6850 printer model

Offset	Type	Size (in bytes)	Description
0	WORD	2	Density LOW ranges from 0 to 250

6800 printer model

Offset	Type	Size (in bytes)	Description
0	WORD	2	Density LOW ranges from 0 to 180

Tag identifier: kPINFTAG_DENSITY_LAMI
Density LAMI setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Density LAMI ranges from 0 to 221

Tag identifier: kPINFTAG_SIDEOFFSET
Side offset setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Side offset ranges from -24 to 24

Tag identifier: kPINFTAG_TOPOFFSET
Top Offset setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Top offset ranges from -24 to 24

Tag identifier: kPINFTAG_SENSOROFFSET
Sensor Offset setting Information

Offset	Type	Size (in bytes)	Description
0	WORD	2	Sensor offset ranges from -24 to 24

kPINFTAG_SIDEOFFSET, kPINFTAG_TOPOFFSET, and
kPINFTAG_SENSOROFFSET
are supported from the following printer firmware versions.

Model	Supported version(main control program version)
6800 Type 1	Version 196 or a later version.
6800 Type 2	Version 309 or a later version.
6800 Type 3	Version 409 or a later version.
6850	Version 614 or a later version.

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_ENABLED
Sensor Offset setting Information

Offset	Type	Size (in bytes)	Description
1	WORD	1	Energy save mode error enabling/disabling. - Disable is 0, enable is 1.

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_AWAKE
Sensor Offset setting Information

Offset	Type	Size (in bytes)	Description
-	-	-	Wake printer up from energy save mode.

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_SLEEP

Sensor Offset setting Information

Offset	Type	Size (in bytes)	Description
-	-	-	Put printer into energy save mode.

Tag identifier: kPINFTAG_ENERGY_SAVE_MODE_WAIT_TIME

Sensor Offset setting Information

Offset	Type	Size (in bytes)	Description
1	BYTE	1	Wait time [min] to enter energy save mode.

rLen Pointer to a double-word type variable that stores the size of data (in bytes)

rResult Pointer to a word type variable that stores the executing status

Output

Return value Returns "TRUE" when the function ends normally or "FALSE" when an undefined tag identifier is specified or when the communication with the printer is disabled.

rResult Executing status

The firmware version of the printer whose "New executing status" corresponds is as follows:

Model	Supported version(main control program version)
6800 Type 1	Version 237 or a newer version.
6800 Type 2	Version 337 or a newer version.
6800 Type 3	Version 437 or a newer version.
6850	Version 657 or a newer version.

(New executing status)

Status	Value	Description
kRESULT_NOERROR	0	No error
kRESULT_PRMERORR	1006	Parameter error such as Undefined Tag and Insufficient Data
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_UNKNOWNPAPER	2301	The specified paper information is not available for the currently selected printer.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (Example: The printer is not powered on.)
kRESULT_BUSY	2100	Now printing
kRESULT_WAIT_WindingRibbonErr	2202	Now rewinding the ink ribbon
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at power-on time.
kRESULT_STATUS_CoverOpenErr	2401	Cover Open.
kRESULT_STATUS_PaperCoverOpenErr	2402	Paper Cover Open.
kRESULT_STATUS_IncorrectRibbonErr	2403	Incorrect Ribbon.
kRESULT_STATUS_IncorrectPaperErr	2404	Incorrect Paper.
kRESULT_STATUS_RibbonEmptyErr	2405	Ribbon Empty.
kRESULT_STATUS_PaperEmptyErr	2408	Paper Empty.
kRESULT_STATUS_ReadyLoadingErr	2412	Ready Loading.
kRESULT_STATUS_TakeOutPaperErr	2413	Take Out Paper.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

The firmware version of the printer whose "Old executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version of 236 or older version.
6800 Type 2	Version of 336 or older version.
6800 Type 3	Version of 436 or older version.
6850	Version of 656 or older version.

(Old executing status)

Status	Value	Description
kRESULT_NOERROR	0	No error
kRESULT_PRMEERROR	1006	Parameter error such as Undefined Tag and Insufficient Data
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_UNKNOWNPAPER	2301	The specified paper information is not available to the currently selected printer.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (Example: The printer is not powered on.)
kRESULT_BUSY	2100	Now printing
kRESULT_WAIT_WindingRibbonErr	2202	Now taking up the ink ribbon
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_OPECALL_PaperEmptyErr	2305	No Paper.
kRESULT_OPECALL_InkEmptyErr	2306	No Ink Ribbon.
kRESULT_OPECALL_UpperCoverOpenErr	2309	Cover Open.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

Remarks

Before executing this function, execute the "chcusb_status" or "chcusb_statusAll" function to check the printer status and make sure that the printer is ready to print (that "rResult" is "kRESULT_READY").

Function name
chcusb_imageformat

Format
BOOL APIENTRY chcusb_imageformat(WORD format, WORD ncomp, WORD depth, WORD width,
WORD height, WORD *rResult);

Function
Specifies the size of an image to be printed.

Input
format Specify the format of an image to be transferred.
The following image formats can be specified.

Format	Value	Description
kFORMAT_PIXEL_RGB	3	RGB point sequential format
kFORMAT_PIXEL_BGR	4	BGR point sequential format

ncomp Specify the number of color components.
The following values can be specified:

Format	Value	Description
kCOMPONENT_RGB	3	Red, green, and blue colors (3 colors)

depth Currently specify "8" as the depth.

width Specify the width of the image in pixels.

Height Specify the height of the image in pixels (in lines).
rResult Pointer to a word type variable for storing the executing status

Output
Return value Returns "TRUE" when the function ends normally or "FALSE" in the other cases.

rResult The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_PRMEERROR	1006	The specified image size cannot be processed (e.g. when "0,0" is specified as the image size).
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Function name
chcusb_setmtf

Format
void APIENTRY chcusb_setmtf(long *mtf);

Function
Specifies parameters for the profile emphasis/smoothing function and sets enabling or disabling of the function.

Input

Mtf

Pointer to the long-type 3×3 array (containing 9 parameters)
Array elements are related to parameters as shown below.

A11 → mtf[0]

A12 → mtf[1]

A13 → mtf[2]

A21 → mtf[3]

A22 → mtf[4]

A23 → mtf[5]

A31 → mtf[6]

A32 → mtf[7]

A33 → mtf[8]

To use (enable) the profile emphasis/smoothing function, specify a parameter value "mtf." If you do not use (disable) the profile emphasis/smoothing function, specify "NULL" for "mtf."
(Specify a value multiplying a value of +10.00 to -10.00 (0.01 step) by 100 for A11 to A33.)

Output

None

Remarks

When the profile emphasis/smoothing function is enabled by this function, the profile of the image output by the "chcusb_write" function is emphasized or smoothed.

Notes

To use the profile emphasis/smoothing function with this function, the "writeSize" specified by the "chcusb_write" function must be a multiple of "width × ncomp" specified by the "chcusb_imageformat" function. If not, the operation of this function is not assured.

Function name
chcusb_makeGamma

Format
 void APIENTRY chcusb_makeGamma(WORD k, BYTE *intoneR, BYTE *intoneG, BYTE *intoneB);

Function
 Creates tone curve data to be used by the "chcusb_setlcctable" function.

Input

k	Value multiplying a gamma coefficient (3.00 to 0.33 at intervals of 0.01) by 100
intoneR	Pointer to a byte type 256-element array for the tone curve R
intoneG	Pointer to a byte type 256-element array for the tone curve G
intoneB	Pointer to a byte type 256-element array for the tone curve B To use a different tone curve for each color, specify NULL for pointers excluding the pointer for the target color. Repeat this for each target color.

Output

IntoneR	Tone curve R calculated by coefficient k
intoneG	Tone curve G calculated by coefficient k
intoneB	Tone curve B calculated by coefficient k

Remarks
 To use a gamma correction coefficient for a tone curve specified by the "chcusb_setlcctable" function, create the table using this function.

Function name
chcusb_setlcctable

Format
BOOL APIENTRY
chcusb_setlcctable(char *inProfileName, char *outProfileName, WORD renderingIntents,
BYTE *inToneR, BYTE *inToneG, BYTE *inToneB,
BYTE *outToneR, BYTE *outToneG, BYTE *outToneB, WORD *rResult);

Function
Enables or disables the color matching function.

Input

inProfileName Specify the path to the input profile.

outProfileName Specify the path to the output profile.
To enable the color matching function, specify the paths to the input and output profiles. To disable the color matching function, specify "NULL" for "inProfileName" and "outProfileName."

renderingIntents Specify one of the following as a matching option:

Constant definition	Value	Description
kICC_RI_Perceptual	0	Photo (Perceptual)
kICC_RI_RelativeColorimetric	1	Logo design (Relative Colorimetric)
kICC_RI_Saturation	2	Chroma saturation (Saturation)

inToneR Pointer to the byte type 256-element array for the input tone curve R

inToneG Pointer to the byte type 256-element array for the input tone curve G

inToneB Pointer to the byte type 256-element array for the input tone curve B

outToneR Pointer to the byte type 256-element array for the output tone curve R

outToneG Pointer to the byte type 256-element array for the output tone curve G

outToneB Pointer to the byte type 256-element array for the output tone curve B

The tone curve is formatted as shown below.

Offset	Type	Size (in bytes)	Description
0	BYTE	1	Output value for the input value 0 (0 to 255)
1	BYTE	1	Output value for the input value 1 (0 to 255)
2	BYTE	1	Output value for the input value 2 (0 to 255)
...	...	...	...
255	BYTE	1	Output value for the input value 255 (0 to 255)

rResult Pointer to the word type variable for storing the executing status

Output

Return value

Returns "TRUE" when the color matching function is enabled with valid I/O profile values or when "NULL" is specified for "inProfileName" and "outProfileName" or "FALSE" when the specified profile is not available or when a memory allocation error occurs.

rResult

Executing status

The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_FILEACCESS	1005	Cannot access the specified profile.
kRESULT_PRMEERROR	1006	The specified profile is not available.
kRESULT_MEMFULLER R	1000	Memory allocation failed.
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Remarks

When the color matching function is enabled, the "chcusb_write" function performs color matching on the image data using the specified profile and sends the result to the printer.

Function name
chcusb_copies

Format
BOOL APIENTRY chcusb_copies(WORD copies, WORD *rResult);

Function
Specifies the number of copies to print.
One image (page) output by the "chcusb_startpage," "chcusb_write" and "chcusb_endpage" functions is printed for the number of times specified by this function.

Input
rResult Pointer to the word type variable for storing the executing status

copies Specify the number of copies to print (1 to 999).

Output
Return value Returns "TRUE" when the "copies" value is 1 to 999 or "FALSE" when the "copies" value is anything else.

rResult Executing status
The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_PRMEERROR	1006	The specified parameter is not available.
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Remarks
The "copies" value set by this function is reset to "1" when the "chcusb_endpage" function ends the transfer of image data.

Function name
chcusb_status

Format
BOOL APIENTRY chcusb_status(WORD *rResult);

Function
Gets the status of the currently selected printer.

Input
rResult Pointer to the word type variable for storing the executing status

Output
Return value Returns "TRUE" when the printer is ready to print or "FALSE" in the other cases.

rResult Executing status

The firmware version of the printer whose "New executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version 237 or a newer version.
6800 Type 2	Version 337 or a newer version.
6800 Type 3	Version 437 or a newer version.
6850	Version 657 or a newer version.

(New executing status)

Status	Value	Description
kRESULT_READY	0	The printer is ready.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread..
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_STATUS_CoverOpenErr	2401	Cover Open.
kRESULT_STATUS_PaperCoverOpenErr	2402	Paper Cover Open.
kRESULT_STATUS_IncorrectRibbonErr	2403	Incorrect Ribbon.
kRESULT_STATUS_IncorrectPaperErr	2404	Incorrect Paper.
kRESULT_STATUS_RibbonEmptyErr	2405	Ribbon Empty.
kRESULT_STATUS_PaperEmptyErr	2408	Paper Empty.
kRESULT_STATUS_ReadyLoadingErr	2412	Ready Loading.
kRESULT_STATUS_TakeOutPaperErr	2413	Take Out Paper.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error
kRESULT_USB_CommErr	2600	USB communication error
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

The firmware version of the printer whose "Old executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version of 236 or older version.
6800 Type 2	Version of 336 or older version.
6800 Type 3	Version of 436 or older version.
6850	Version of 656 or older version.

(Old executing status)

Status	Value	Description
kRESULT_READY	0	The printer is ready.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread..
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_OPECALL_PaperEmptyErr	2305	No Paper.
kRESULT_OPECALL_InkEmptyErr	2306	No Ink Ribbon.
kRESULT_OPECALL_UpperCoverOpen Err	2309	Cover Open.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

Function name
chcusb_statusAll

Format
BOOL APIENTRY plugin_statusAll(BYTE *idArray, WORD *rResultArray);

Function
Acquires the printer status for all of the printers ID's in the idArray parameter

Input
IdArray Printer ID number list created with the listupPrinter function (pointer to the 128-element BYTE type array)

rResultArray Pointer to the 128-element word type array that stores printer status)

Output
Return value Returns "TRUE" even when there is just one printer available for printing.
Returns "FALSE" when there is no printer available for printing

rResultArray The status code values are the values returned by rResult (the output of the chcusb status function). The number of status code values corresponds to the number of printers specified in idArray. The status code corresponding to the printer specified as idArray[0] is rResultArray[0], and status code corresponding to idArray[1] is rResultArray[1].)

The firmware version of the printer whose "New executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version 237 or a newer version.
6800 Type 2	Version 337 or a newer version.
6800 Type 3	Version 437 or a newer version.
6850	Version 657 or a newer version.

(New executing status)

Status	Value	Description
kRESULT_READY	0	The printer is ready.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread..
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_STATUS_CoverOpenErr	2401	Cover Open.
kRESULT_STATUS_PaperCoverOpenErr	2402	Paper Cover Open.
kRESULT_STATUS_IncorrectRibbonErr	2403	Incorrect Ribbon.
kRESULT_STATUS_IncorrectPaperErr	2404	Incorrect Paper.
kRESULT_STATUS_RibbonEmptyErr	2405	Ribbon Empty.
kRESULT_STATUS_PaperEmptyErr	2408	Paper Empty.
kRESULT_STATUS_ReadyLoadingErr	2412	Ready Loading.
kRESULT_STATUS_TakeOutPaperErr	2413	Take Out Paper.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

The firmware version of the printer whose "Old executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version of 236 or older version.
6800 Type 2	Version of 336 or older version.
6800 Type 3	Version of 436 or older version.
6850	Version of 656 or older version.

(Old executing status)

Status	Value	Description
kRESULT_READY	0	The printer is ready.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread..
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_OPECALL_PaperEmptyErr	2305	No Paper.
kRESULT_OPECALL_InkEmptyErr	2306	No Ink Ribbon.
kRESULT_OPECALL_UpperCoverOpen Err	2309	Cover Open.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

Function name
chcusb_startpage

Format
BOOL APIENTRY chcusb_startpage(WORD* pageID, WORD *rResult);

Function
Sends a printer command to initiate printing a page on the printer.

Input
PageID Pointer to word type variable that stores page ID.

rResult Pointer to the word type variable for storing the executing status

Output
Return value Returns "TRUE" when the function ends normally or "FALSE" in the other case.

pageID Page ID of the image data to be sent. After the image data is sent, you can clear the number of copies to print, and browse the number of copies left, by using this page ID

The page ID is formatted as shown below.

Higher order byte		Lower order byte	
B15	B8	B7	B0
Printer ID number(0~127)		Number assigned inside DLL nominally 1~255	

When a multiple number of printers is connected to the computer, you can identify the printer page ID by looking at the lower order byte only.

rResult Executing status

The firmware version of the printer whose "New executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version 237 or a newer version.
6800 Type 2	Version 337 or a newer version.
6800 Type 3	Version 437 or a newer version.
6850	Version 657 or a newer version.

(New executing status)

Status	Value	Description
kRESULT_NOERROR	0	No error
kRESULT_PRMERROR	1006	Parameter error The picture size is not available to the printer.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_STATUS_CoverOpenErr	2401	Cover Open.
kRESULT_STATUS_PaperCoverOpenErr	2402	Paper Cover Open.
kRESULT_STATUS_IncorrectRibbonErr	2403	Incorrect Ribbon.
kRESULT_STATUS_IncorrectPaperErr	2404	Incorrect Paper.
kRESULT_STATUS_RibbonEmptyErr	2405	Ribbon Empty.
kRESULT_STATUS_PaperEmptyErr	2408	Paper Empty.
kRESULT_STATUS_ReadyLoadingErr	2412	Ready Loading.
kRESULT_STATUS_TakeOutPaperErr	2413	Take Out Paper.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCC		

The firmware version of the printer whose "Old executing status" corresponds is as follows.

Model	Supported version(main control program version)
6800 Type 1	Version of 236 or older version.
6800 Type 2	Version of 336 or older version.
6800 Type 3	Version of 436 or older version.
6850	Version of 656 or older version.

(Old executing status)

Status	Value	Description
kRESULT_NOERROR	0	No error
kRESULT_PRMERROR	1006	Parameter error The picture size is not available to the printer.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized. (The printer is not powered on.)
kRESULT_BUSY	2100	The printer is busy in printing.
kRESULT_WAIT_WindingRibbonErr	2202	The ink ribbon is being taken up.
kRESULT_WAIT_OffLineErr	2204	The operator panel is being used.
kRESULT_WAIT_InitializeErr	2205	The printer is initializing at the power-on time.
kRESULT_OPECALL_PaperEmptyErr	2305	No Paper.
kRESULT_OPECALL_InkEmptyErr	2306	No Ink Ribbon.
kRESULT_OPECALL_UpperCoverOpenErr	2309	Cover Open.
kRESULT_SVCCALL_UNKOWN	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communication error.
kRESULT_OPECALL_PaperJamErr	Refer to KODAK 68xx Printer Error Code List.	
kRESULT_SVCCALL		

Function name
chcusb_endpage

Format
BOOL APIENTRY chcusb_endpage(WORD *rResult);

Function
Performs the page end processing.

Input
rResult Pointer to the word type variable for storing the executing status

Output
Return value This function always returns "TRUE."

rResult Executing status

The following states can be stored:

Status	Value	Description
kRESULT_NOERROR	0	No error
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Function name
chcusb_write

Format
BOOL APIENTRY chcusb_write(BYTE *data, DWORD *writeSize, WORD *rResult);

Function
Sends image data to the printer.

Input

data	Pointer to the output image data
writeSize	Pointer to the double-word type variable that stores the output size
rResult	Pointer to the word type variable for storing the executing status

Output

Return value	Returns "TRUE" when the function ends normally or "FALSE" in the other cases.
writeSize	Stores the output data size (in bytes).
rResult	Executing status

The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized.
kRESULT_USB_CommErr	2600	USB communication error

A status value which is equal to the executing status of the "chcusb_status" function may be returned if this function is called when the printer is in an error status ("kRESULT_NOERROR" by the "chcusb_startpage" function is not checked) or if the data transmitted is greater than the size set by the "chcusb_imageformat" function.

Remarks

- When the profile emphasis/smoothing function is enabled (ON) by the "chcusb_setmtf" function, the image data is MTF-processed in this function before being transmitted to the printer.
- When the color matching parameter is set in the "chcusb_setlcctable" function, the image data is color-matching-processed in this function before being transmitted to the printer.
- When the tone-curve parameter is set in the "chcusb_setlcctable" function, the image data is tone-curve-processed in this function before being transmitted to the printer.

Notes:

The input parameter "writeSize" must be a multiple of the image "width × ncomp" specified by the "chcusb_imageformat" function. If it is not a multiple of "width × ncomp," the functions set by the "chcusb_setmtf" function and the "chcusb_setlcctable" function are disabled.

Function name
chcusb_getGamma

Format
BOOL APIENTRY chcusb_getGamma(char *filename, BYTE *r, BYTE *g, BYTE *b, WORD *rResult);

Function
Loads tone curve data from a file.

Input
filename

Path to the tone curve data file
The tone curve data file is formatted as follows:

Offset	Type	Size	Description
0	BYTE	1	Output value for the R input value 0 (0 to 255)
1	BYTE	1	Output value for the R input value 1 (0 to 255)
2	BYTE	1	Output value for the R input value 2 (0 to 255)
:	:	:	:
255	BYTE	1	Output value for the R input value 255 (0 to 255)
256	BYTE	1	Output value for the G input value 0 (0 to 255)
257	BYTE	1	Output value for the G input value 1 (0 to 255)
258	BYTE	1	Output value for the G input value 2 (0 to 255)
:	:	:	:
511	BYTE	1	Output value for the G input value 255 (0 to 255)
512	BYTE	1	Output value for the B input value 0 (0 to 255)
513	BYTE	1	Output value for the B input value 1 (0 to 255)
514	BYTE	1	Output value for the B input value 2 (0 to 255)
:	:	:	:
767	BYTE	1	Output value for the B input value 255 (0 to 255)

r Pointer to the word type 256-byte array that stores the R tone curve

g Pointer to the word type 256-byte array that stores the G tone curve

b Pointer to the word type 256-byte array that stores the B tone curve

rResult Pointer to the word type variable that stores the executing status

Output

Return value Returns "TRUE" when the tone curve is read normally or "FALSE" in the other cases.

r R (Red) tone curve data

g G (Green) tone curve data

b B (Blue) tone curve data

rResult Executing status

The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_FILEACCESS	1005	Cannot access the file or cannot read the data.
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Function name
chcusb_getMtf

Format
BOOL APIENTRY chcusb_getMtf(char *filename, long *mtf, WORD *rResult);

Function
Reads parameters of the profile emphasis/smoothing function (MTF) from the file.

Input
filename Path to the MTF parameter file
The MTF parameter file is a text file containing 9 values (in an array of 3 elements by 3 elements) that are separated from each other by a space. The parameters are arrayed as follows:

A11 A12 A13

A21 A22 A23

A31 A32 A33

(A11 to A33 are values multiplying +10.00 to -10.00 by 100.)

Two examples of MTF parameter files are given below. (These examples are functionally the same.)

- Example 1:
0 0 0
0 100 0
0 0 0
- Example 2:
0 0 0 0 100 0 0 0 0

mtf Pointer to a long type 9-element array (3 elements by 3 elements)

rResult Pointer to the word type variable that stores the executing status

Output
Return value Returns "TRUE" when the MTF parameters are read normally or "FALSE" in the other cases.

mtf MTF parameters that are read

rResult Executing status
The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_FILEACCESS	1005	Cannot access the specified profile or cannot read the data.
kRESULT_THREAD_Max	1501	Cannot generate new thread.

Remarks

You can use the MTF function by transferring the MTF parameters obtained by this function to the "chcusb_setMtfFile" function. If the application need not browse the MTF parameters, you can read and apply the parameters by using the "chcusb_setMtfFile" function.

Function name
chcusb_cancelCopies

Format
BOOL APIENTRY chcusb_cancelCopies (WORD *pageID, WORD *rResult);

Function
Cancels jobs for image data already sent with a specified page ID, by re-setting the number of copies for the print.

Input
pageID Page ID which cancels jobs.(page ID acquired with chcusb_startpage function)

When 0xFFFFh is specified, both the image in process of printing by the current printer, and the second image which is received by the buffer, will be cancelled

rResult Pointer to the word type variable that stores the executing status

Output
Return value Returns "TRUE" when the function ends normally or "FALSE" in the other cases.

rResult Executing status
The states to be stored are as follows:

Status	Value	Description
kRESULT_NOERROR	0	Normal ("TRUE" is returned.)
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	The printer cannot be recognized.
kRESULT_USB_CommErr	2600	USB communication error
kRESULT_BUSY	2100	The printer is receiving data and the job cannot be canceled.

Remarks

This function is not valid for the image page that is currently being printed.
This function cannot be used while the printer is receiving the next image data.
When this command is executed, note that the number of prints left, the number of completed prints, and the specified print count number for the specified page ID shall be all cleared.

Function name

chcusb_setPrinterToneCurve

Format

BOOL APIENTRY chcusb_setPrinterToneCurve (WORD type, WORD *data, WORD *rResult);

Function

Changes tone curve table inside the printer.

Input

Type

Sets the printer tone curve type.

The following kind of printer tone curves can be specified.

Tag	Value	Description
kSetPrinterTCTbl_StandardTbl	0	Changes to the standard tone curve setting inside printer
KSetPrinterTCTbl_UserTbl	1	Set up the user tone curve
kSetPrinterTCTbl_CurrentTbl	2	Set up the current tone curve

data

Pointer for the 768-element (YMC each 256 elements) tone curve table word type array.

If kPrinterToneCurve_StandardTbl is selected, it's set up NULL.

Tone curve table is formatted as follows:

Offset	Type	Size	Description
0	WORD	2	Output value (0~16383) for Yellow input value 0
2	WORD	2	Output value (0~16383) for Yellow input value 1
:	:	:	:
510	WORD	2	Output value (0~16383) for Yellow input value 255
512	WORD	2	Output value (0~16383) for Magenta input value 0
514	WORD	2	Output value (0~16383) for Magenta input value 1)
:	:	:	:
1020	WORD	2	Output value (0~16383) for Magenta input value 255
1022	WORD	2	Output value (0~16383) for Cyan input value 0
1024	WORD	2	Output value (0~16383) for Cyan input value 1
:	:	:	:
1530	WORD	2	Output value (0~16383) for Cyan input value 255

rResult

Pointer to the word type variable that stores the Executing status.

Output

Return value

Returns "TRUE" when the function successfully closes, or "FALSE" in all other cases.

rResult

Executing status

The states to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Normal ("TRUE" is returned)
kRESULT_PRMEERROR	1006	Parameter error such as Undefined Tag and Insufficient Data
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Cannot recognize the printer
kRESULT_USB_CommErr	2600	USB communication error.

Remarks

The Tone Curve Types are as follows:

- Standard Tone Curve
Standard Tone Curve is contained within the printer firmware.
It is stored inside flash memory, and users cannot re-write it to make changes.
The only way to re-write it is to download different firmware.
- User Tone Curve
User Tone Curve is stored inside the printer's flash memory.
Users can re-write it by downloading new values to the printer.
Since it is stored within the flash memory area inside the printer, the values are retained even when the power is turned off.
- Current Tone Curve
Current Tone Curve is stored in RAM, and is used at time of printing.
Users can re-write it by downloading new values to the printer.
However, the contents are deleted when the printer power is turned off.

Please refer to appendix B for a detailed explanation.

Function name

chcusb_setPrinterToneCurve2

Format

```
BOOL APIENTRY chcusb_setPrinterToneCurve2( WORD type, WORD number, WORD *data,
WORD *rResult );
```

Function

Changes the internal tone curve table of the printer.

Parameters

[IN]

Type

Specify the desired type of printer tone curve.

The types of printer tone curve tables that can be specified are listed below.

Tone curve type	Value	Description
kSetPrinterTCTbl_StandardTbl	0	Re-set to standard tone curve.
kSetPrinterTCTbl_UserTbl	1	Set the user tone curve
kSetPrinterTCTbl_CurrentTbl	2	Set the current tone curve.

[IN]

number

Select the printer tone curve number.

Selectable tone curve table numbers are as follows.

Tone curve table numbers	Value	Donor Code
kSetPrinterTCTbl_ToneTable1	0	6x8" 6SR
kSetPrinterTCTbl_ToneTable2	1	6x8" 6PR 5x7" 5PR
kSetPrinterTCTbl_ToneTable3	2	6x8" 6TR2 5x7" 5TR2 6x8" 6TR Reserve1 5x7" 5TR Reserve1 6x8" 6TR Reserve2 5x7" 5TR Reserve2 6x8" 6TR Reserve3 5x7" 5TR Reserve3

[IN]

data

Pointer to the WORD-type array of 768 tone curve table elements
(256 elements for Y, M, and C each)

The format of the tone curve tables is shown below.

Offset	Type	Size (in bytes)	Description
0	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for yellow
2	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for yellow
:	:	:	:
510	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for yellow
512	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for magenta
514	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for magenta
:	:	:	:
1022	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for magenta
1024	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for cyan
1026	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for cyan
:	:	:	:
1534	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for cyan

[IN/OUT]

rResult Pointer to the WORD-type variables for execution status storage

Output

Return value Returns TRUE when this function is executed to completion, or returns FALSE in all other cases.

rResult Execution status
The types of status information stored are listed below.

Status	Value	Meaning
kRESULT_NOERROR	0	Normal (TRUE is assigned as the return value)
kRESULT_PRMEERROR	1006	Parameter error. Caused by: failure to define the tag, the insufficiency in data size, or others.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	The created thread is not usable.
kRESULT_NOTCONNECT	2000	The printer cannot be identified.
kRESULT_STATUS_BUSY	2100	Printer is busy.
kRESULT_Unknown	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communications error.

Remarks

(1)

To use the same parameter each time for printing, set the "chcusb_setPrinterToneCurve" function after executing the "chcusb_open" function, since the "chcusb_setPrinterToneCurve" function, once set, will remain in memory until the "chcusb_close" function is executed.

- Standard tone curve
Tone curves that is built-in to the printer firmware.
That is recorded in the flash memory, and it would not be possible to change by user.
It is changeable by changing firmware.
- Current tone curve
Tone curve that is used on printing.
It is recorded on to RAM, and it is changeable by download to the printer by user.
However when turned power off, the specified contents would be canceled.
- User' tone curve
Tone curve that is recorded in the flash memory of the printer
It is changeable by download to the printer by user.
As it is recorded in the area of flash memory, the specified contents would be recorded when turned power off.

(2)

The version which the printer of each model supports this function is as follows;

Model	Supported version(main control program version)
6800 Type 1	234
6800 Type 2	334
6800 Type 3	434
6850	654

Function name
chcusb_getPrinterToneCurve

Format
BOOL APIENTRY chcusb_getPrinterToneCurve (WORD type, WORD *state, WORD *data, WORD *rResult);

Function
Acquires tone curve table inside the printer.

Input
Type Sets the printer tone curve type.
The following kind of tone curves can be specified.

Tag	Value	Description
kGetPrinterTCTbl_UserTbl	1	Acquires user tone curve table
kGetPrinterTCTbl_CurrentTbl	2	Acquires current tone curve table

rResult Pointer to the word type variable that stores Executing status.

Output
state The states to be stored are as follows:

Offset	Type	Size	Description
0	WORD	2	Output the present tone curve type. 00h...Used to select standard tone curve table 01h...Used to select user tone curve table 02h...Used to select current tone curve table

data Pointer for the 768-element (YMC each 256 elements) tone curve table word type array.
Tone curve table is formatted as follows:

Offset	Type	Size	Description
0	WORD	2	Output value (0~16383) for Yellow input value 0
2	WORD	2	Output value (0~16383) for Yellow input value 1
:	:	:	:
510	WORD	2	Output value (0~16383) for Yellow input value 255
512	WORD	2	Output value (0~16383) for Magenta input value 0
514	WORD	2	Output value (0~16383) for Magenta input value 1)
:	:	:	:
1020	WORD	2	Output value (0~16383) for Magenta input value 255
1022	WORD	2	Output value (0~16383) for Cyan input value 0
1024	WORD	2	Output value (0~16383) for Cyan input value 1
:	:	:	:
1530	WORD	2	Output value (0~16383) for Cyan input value 255

rResult

Executing status

The states to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Normal ("TRUE" is returned)
kRESULT_PRMERROR	1006	Parameter error such as Undefined
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Cannot recognize the printer
kRESULT_USB_CommErr	2600	USB communication error.

Remarks

The Tone Curve Types are as follows:

- Standard Tone Curve
Standard Tone Curve is contained within the printer firmware.
It is stored inside flash memory, and users cannot re-write it to make changes.
The only way to re-write it is to download different firmware.
- User Tone Curve
User Tone Curve is stored inside the printer's flash memory.
Users can re-write it by downloading new values to the printer.
Since it is stored within the flash memory area inside the printer, the values are retained even when the power is turned off.
- Current Tone Curve
Current Tone Curve is stored in RAM, and is used at time of printing.
Users can re-write it by downloading new values to the printer.
However, the contents are deleted when the printer power is turned off.

Please refer to appendix B for a detailed explanation.

Function name

chcusb_getPrinterToneCurve2

Format

```
BOOL APIENTRY chcusb_getPrinterToneCurve2( WORD type, WORD number, WORD *data,
WORD *rResult );
```

Function

Acquires the internal tone curve table of the printer.

Parameters

[IN]

Type

Specify the desired type of printer tone curve.

The types of printer tone curve tables that can be specified are listed below.

Tone curve type	Value	Description
kGetPrinterTCTbl_UserTbl	1	Set the user tone curve
kGetPrinterTCTbl_CurrentTbl	2	Set the current tone curve.

[IN]

number

Select the printer tone curve number.

Selectable tone curve table numbers are as follows.

Tone curve table numbers	Value	Donor Code
kGetPrinterTCTbl_ToneTable1	0	6x8" 6SR
kGetPrinterTCTbl_ToneTable2	1	6x8" 6PR 5x7" 5PR
kGetPrinterTCTbl_ToneTable3	2	6x8" 6TR2 5x7" 5TR2 6x8" 6TR Reserve1 5x7" 5TR Reserve1 6x8" 6TR Reserve2 5x7" 5TR Reserve2 6x8" 6TR Reserve3 5x7" 5TR Reserve3

[OUT]

data

Pointer to the WORD-type array of 768 tone curve table elements
(256 elements for Y, M, and C each)

The format of the tone curve tables is shown below.

Offset	Type	Size (in bytes)	Description
0	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for yellow
2	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for yellow
:	:	:	:
510	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for yellow
512	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for magenta
514	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for magenta
:	:	:	:
1022	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for magenta
1024	BIN	2	Output value (0 - 16383) with respect to an input value of 0 for cyan
1026	BIN	2	Output value (0 - 16383) with respect to an input value of 1 for cyan
:	:	:	:
1534	BIN	2	Output value (0 - 16383) with respect to an input value of 255 for cyan

[IN/OUT]

rResult Pointer to the WORD-type variables for execution status storage

Output

Return value Returns TRUE when this function is executed to completion, or returns FALSE in all other cases.

rResult Execution status
The types of status information stored are listed below.

Status	Value	Meaning
kRESULT_NOERROR	0	Normal (TRUE is assigned as the return value)
kRESULT_PRMEERROR	1006	Parameter error. Caused by: failure to define the tag, the insufficiency in data size, or others.
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	The created thread is not usable.
kRESULT_NOTCONNECT	2000	The printer cannot be identified.
kRESULT_STATUS_BUSY	2100	Printer is busy.
kRESULT_Unknown	2504	Undefined error.
kRESULT_USB_CommErr	2600	USB communications error.

Remarks

(1)

- Standard tone curve
Tone curves that is built-in to the printer firmware.
That is recorded in the flash memory, and it would not be possible to change by user.
It is changeable by changing firmware.
- Current tone curve
Tone curve that is used on printing.
It is recorded on to RAM, and it is changeable by download to the printer by user.
However when turned power off, the specified contents would be canceled.
- User' tone curve
Tone curve that is recorded in the flash memory of the printer
It is changeable by download to the printer by user.
As it is recorded in the area of flash memory, the specified contents would be recorded when turned power off.

(2)

The version which the printer of each model supports this function is as follows;

Model	Supported version(main control program version)
6800 Type 1	234
6800 Type 2	334
6800 Type 3	434
6850	654

Function name

chcusb_blinkLED

Format

BOOL APIENTRY chcusb_blinkLED (WORD *rResult);

Function

Flashes LED of the current selected printer for 5 times

Input

rResult Pointer to the word type variable that stores Executing status.

Output

Return value Returns "TRUE" when the function successfully closes, or "FALSE" in all other cases.

rResult

Executing status

The states to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Normal ("TRUE" is returned)
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Cannot recognize the printer
kRESULT_USB_CommErr	2600	USB communication error.

Function name

chcusb_resetPrinter

Format

BOOL APIENTRY chcusb_resetPrinter (WORD *rResult);

Function

Resets the currently selected printer.

Input

rResult Pointer to the word type variable that stores Executing status.

Output

Return value Returns "TRUE" when the function successfully closes, or "FALSE" in all other cases.

rResult

Executing status

The states to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Normal("TRUE" is returned)
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_NOTCONNECT	2000	Cannot recognize the printer
kRESULT_USB_CommErr	2600	USB communication error.

Remarks

If this function is implemented while printing, the printer is reset when printing is finished.

If the number of copies was set using the plugin_copies function, the number of copies is cancelled and the printer is reset after completing the present print.

This function becomes a wait condition until the reset action is complete.

Function name

chcusb_attachThreadCount

Format

BOOL APIENTRY chcusb_attachThreadCount(WORD*rCount, WORD*rMaxCount);

Function

Checks the maximum number of threads requested, and the number of threads that are currently being used.

Input

rCount Pointer to the word type variable that stores the number of threads currently being used.

rMaxCount Pointer to the word type variable that stores the maximum number of threads requested.

Output

rCount Returns the current thread count.

rMaxCount Returns the maximum number of threads requested by the chcusb_makeThread function.

Remarks

If the chcusb_makeThread function is not used, the maximum thread number returned is 1.

Function name

chcusb_getColorSensorLevel

Format

BOOL APIENTRY chcusb_getColorSensorLevel (WORD *rLevels, WORD *rResult);

Function

Acquires color sensor Level of the ribbon set to the printer.

Input

rLevels Pointer to memory that stores color sensor levels.
The size of memory is 24 bytes.

rResult Pointer to the word type variable that stores Executing status.

Output

rLevels Pointer for the 12-element color sensor level table word type array.
The unit of color sensor level is millivolt (mV).
The table is formatted as follows:

Offset	Type	Size	Description
0	WORD	2	Red color sensor voltage on yellow patch of ribbon.
2	WORD	2	Green color sensor voltage on yellow patch of ribbon.
4	WORD	2	Blue color sensor voltage on yellow patch of ribbon.
6	WORD	2	Red color sensor voltage on magenta patch of ribbon.
8	WORD	2	Green color sensor voltage on magenta patch of ribbon.
10	WORD	2	Blue color sensor voltage on magenta patch of ribbon.
12	WORD	2	Red color sensor voltage on cyan patch of ribbon.
14	WORD	2	Green color sensor voltage on cyan patch of ribbon.
16	WORD	2	Blue color sensor voltage on cyan patch of ribbon.
18	WORD	2	Red color sensor voltage on OP patch of ribbon.
20	WORD	2	Green color sensor voltage on OP patch of ribbon.
22	WORD	2	Blue color sensor voltage on OP patch of ribbon.

rResult

Executing status
The states to be stored are as follows:

Status	Value	Meaning
kRESULT_NOERROR	0	Normal ("TRUE" is returned)
kRESULT_MUTEX_Wait	1500	Printer is used by another thread.
kRESULT_THREAD_Max	1501	Cannot generate new thread.
kRESULT_BUSY	2100	The printer is busy or old version firmware.
kRESULT_NOTCONNECT	2000	Cannot recognize the printer
kRESULT_USB_CommErr	2600	USB communication error.

APPENDIX A SOFTWARE DEVELOPMENT KIT

The plug-in DLL development kit contains the following:

- ChcusbDllSdk\SDK folder
This folder contains all files that must be installed in the application folder.
CHCUSB.dll Main body of the CHCUSB DLL
SDKColorMatch.dll Submodule for color matching
SDKUsbCtrl.dll USB access module
CHCUSB.h C header file (function prototype, constant definition, etc.)
CHCUSB.lib IMPORT library for VC++ v6.0 of CHCUSB.DLL
- ChcusbDllSdk\SampleCode folder
This folder contains source files for sample programs using the plug-in DLL and the related files for installation.

APPENDIX B Printer Tone Curve

The printer tone curve function is illustrated in the following diagram.

*The number below corresponds to the number in the figure.

1. The tone curve to be used at time of printing is always the tone curve in RAM.
2. If the user tone curve has never been set from the host PC:
When the printer is turned on, the standard tone curve is copied into the current tone curve (RAM).
3. If the user tone curve has been set from the host PC:
When the printer is turned on, the user tone curve is copied into the current tone curve (RAM).
4. The user tone curve can be re-written from PC, by the setPrinterToneCurve function.
When the user tone curve is re-written, the current tone curve is rewritten with the same contents automatically.
Note:
Even if the current tone curve has already been written from the host PC, it is rewritten by writing the user tone curve.
5. The user tone curve can be read out to the PC, using the getPrinterToneCurve function.
6. The current tone curve can be re-written from the PC using the setPrinterToneCurve function.
The current tone curve data is deleted when printer power is turned off.
7. The standard tone curve can read out to the PC, using the getPrinterToneCurve function.
8. The contents of the current tone curve and user tone curve can be changed to the standard tone curve values using the setPrinterToneCurve function from the host PC.
To do this, only the commands need to be sent, the tone curve data need not be sent.

Printer

